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Richard R. Doell, G. Brent Dalrymple, Robert L. Smith and Roy A. Bailey

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Paleomagnetism, Potassium-Argon Ages, and Geology of Rhyolites and Associated Rocks of the Valles Caldera, New Mexico

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ABSTRACT

Paleomagnetic and potassium-argon studies support geologic evidence that the lower member of the Bandelier Tuff was deposited 1.4 m.y. ago. The upper member erupted about 1.0 m.y. ago and was followed by caldera collapse which formed the 12- to 14-mile diameter Valles Caldera. Post-caldera activity which resulted in the eruption of rhyolite domes and pyroclastic material, has occurred at about 0.9, 0.7, 0.5, and 0.4 m.y. ago, with later undated eruptions that were estimated at about 0.1 m.y. ago.

These data from the Valles Caldera are the basis for the previously published age revision of the Brunhes-Matuyama geomagnetic polarity epoch boundary from 1.0 to 0.7 m.y. ago, and they were used to define the Jaramillo normal polarity event at about 0.9 m.y. ago (Doell and Dalrymple, 1966).

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INTRODUCTION

The Valles Caldera is located in the Jemez Mountains, a complex volcanic highland of Pliocene and Pleistocene age, about 35 miles northwest of Santa Fe in north-central New Mexico (Fig. 1). The highland rises from an eroded central mountainous mass that is composed mainly of basaltic, andesitic and dacitic rocks and is surrounded by more youthfully dissected plateaus that are composed mainly of welded rhyolitic ash flows. In the crest of the high-

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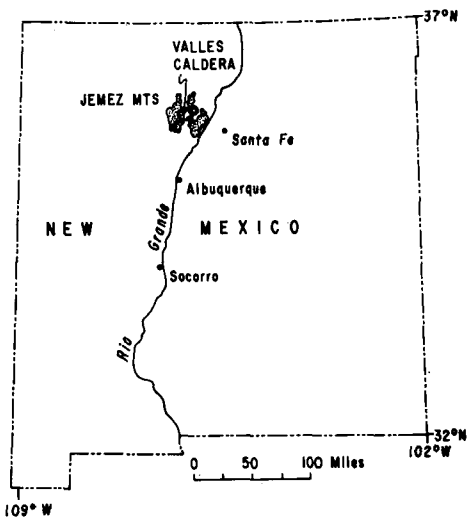


Figure 1. Index map showing location of the Jemez Mountains and the Valles Caldera. Stippling indicates distribution of the Bandelier Tuff.

land is the Valles Caldera, a subcircular depression from 12 to 14 miles in diameter and from 500 to 2000 feet deep. In the center of the caldera, a broad central domical mountain (Redondo)—a structural dome — rises to an elevation of over 11,000 feet. Peripheral to this central structure are several other smaller, but more steep-sided, mountains, which are young rhyolite domes.

The volcanic rocks of the Jemez Mountains overlap the western margin of the Rio Grande graben, a major tectonic feature that extends several hundred miles north-south through New Mexico and southern Colorado. They overlie, on the west, Precambrian granite and Paleozoic and Mesozoic sedimentary rocks and on the east, Tertiary epiclastic sediments that fill the graben.

Volcanism that resulted in the precaldera volcanic complex began in early Pliocene time and continued through late Pliocene with eruption of basaltic, andesitic, dacitic, and rhyolitic lavas and tuffs. About 1.0 and 1.4 m.y. ago, during the Pleistocene, volcanism was climaxed by two eruptive episodes, each of which resulted in deposition of about 50 cubic miles of rhyolite ash and pumice from turbulent ashflows, followed by caldera collapse. The earlier eruptive episode resulted in the deposition of the lower member of the Bandelier Tuff and the formation of the Toledo Caldera, only part of which is now preserved. Postcaldera activity included eruption of rhyolite pyroclastics and lava domes from vents within the caldera. Deposition of the upper member of the Bandelier Tuff and formation of the Valles Caldera took place during the later eruptive episode. Postsubsidence volcanic activity within the Valles Caldera, which perhaps extended up to 100,000 years ago, consisted of structural uplift of the caldera floor that was accompanied and followed by eruption of rhyolites along the caldera-ring fracture; details are presented in other publications (Smith and others, 1961; Ross and others, 1963; Smith and Bailey, 1966, 1968; Smith and others, in press). Further work on the general geology and petrology of the Jemez Mountains is in progress.

The present study concerns paleomagnetic measurements and geochronology of the Bandelier Tuff and the postsubsidence rhyolites within the Valles Caldera. Since these rocks range in age from a few hundred thousand years to about one and one-half million years, they have yielded polarity-age data

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of importance in defining the last, or Brunhes-Matuyama polarity transition of the geomagnetic polarity epoch time scale (Cox and others, 1964; Dalrymple and others, 1965; Doell and others, 1966; McDougall and Chamalaun, 1966) and the Jaramillo normal-polarity event (Doell and Dalrymple, 1966).

STRATIGRAPHY

The sequence of stratigraphic units discussed in this report is shown in the explanation of Plate 1.

Sawyer Dome

Sawyer dome (unit number S29[1]) is a well-preserved volcanic dome of dark-gray hornblende quartz latite on the southeast rim of the Valles Caldera. Geologic evidence for its age is contradictory. Its composition and lithology suggest that it is related to the pre-Toledo Caldera dacites and quartz latites that dominate the northern and higher parts of the Jemez Mountains volcanic pile. However, its youthful morphology and its mineralogic similarity to a unique pair of quartz latite domes known to bracket in time the upper part of the Bandelier Tuff in the Toledo Caldera suggest that it may be younger

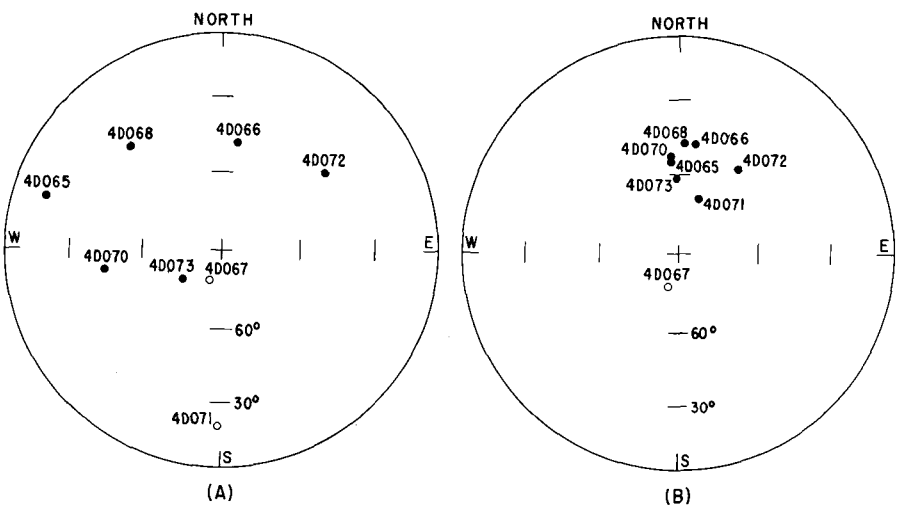


Figure 2. Directions of remanent magnetism in samples collected from San Antonio Mountain II (unit no. S45). (A) Natural remanent magnetization. (B) After partial alternating-field demagnetization at 400 oersteds peak value. Solid symbols on lower hemisphere and open symbols on upper hemisphere of Lambert equal-area projections. Numbers adjacent to points identify individual sample cores.

[1]Each volcanic unit studied in this report that can be considered as a single cooling unit has been assigned a unit number (S29 through S47). Numbers such as 3X111 or 4DO67 refer to individual samples collected from those units.

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106°45'
36°00'
106°30'
35°45'

Warm Spring dome S36
CERRO SECO S39
CERRO SAN LUIS S41
CERRO SANTA ROSA S38
S40
S37
SAN ANTONIO MTN
S45
S33
S30
CERRO DEL ABRIGO
CERRO DEL MEDIO
S34
S35
S42
S32
S31
S46
S47
BANCO BONITO
CERRO LA JARA
S43
SOUTH MTN
S44
SAWYER DOME
S29
Battleship Rock
CANON DE SAN DIEGO
VIRGIN CANYON
CALAVERAS CANYON
RIO CEBOLLA
JEMEZ PLATEAU
PAJARITO PLATEAU
126
4
0 2 4 6 MILES

EXPLANATION

Pleistocene
Intra-Caldera rhyolites
Alluvium and caldera fill
Obsidian flow of Banco Bonito
Ash flows of Battleship Rock
Late rhyolite — Group C, Group B, Group A
Middle rhyolite
Early rhyolite
QUATERNARY
Bandelier Tuff — Upper member, Lower member
Cerro Toledo Rhyolite
Pre-Pleistocene
Pre-caldera volcanic and sedimentary rocks
Includes rocks of Sawyer Dome
PRE-QUATERNARY
Contact
Normal fault
Bar and ball on downthrown side
Topographic rim of Valles Caldera
Topographic rim of Toledo Caldera
S39 Sample locality

GENERALIZED GEOLOGIC MAP OF VALLES CALDERA SHOWING LOCATION OF SAMPLES

DOELL AND OTHERS, PLATE 1
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than the lower part of the Bandelier Tuff. No K-Ar age was determined for this unit, but, as will be discussed later, paleomagnetic measurements suggest an age older than both parts of the Bandelier Tuff.

Bandelier Tuff

The Bandelier Tuff forms gently sloping plateaus that cover 500 square miles on the north, east, and west flanks of the Jemez Mountains. These plateaus are deeply dissected by radial streams, and numerous steep-walled canyons in the tuff provide excellent exposures for sampling and detailed stratigraphic studies.

The formation consists of two major ash-flow members; each member is underlain by a genetically related, bedded air-fall pumice. The names of the ash-flow members were formalized by Griggs (1964) as the Otowi and Tshirege Members. The bedded air-fall pumice beneath the Otowi Member was named the Guaje Member by Griggs, but the air-fall pumice associated with the ash-flow tuff of the Tshirege Member was apparently included by him within the base of the Tshirege Member. In this paper we will follow the usage of Smith and Bailey (1966) and use the informal terms, lower member and upper member. The lower member comprises Grigg's Otowi (an ash-flow unit) and the underlying Guaje Member (the associated air-fall unit), and the upper member constitutes the Tshirege Member, including the basal air-fall pumice. Thus, each of the lower and upper members consist of an ash-flow unit and its associated basal air-fall pumice.

The lower member overlies a marked erosional unconformity that is developed on the older volcanic rocks of the Jemez Mountains. The upper member, in turn, is separated from the lower member by a marked erosional disconformity and locally by bedded tuffs, glowing avalanche deposits, and volcanic sediments that related to the Cerro Toledo Rhyolite of the Toledo Caldera (Pl. 1).

The internal stratigraphy of both members of the Bandelier Tuff is complex. *Each* member is composed of numerous superposed *ash flows,* in which has developed a complex zonation of welding and crystallization that is related to emplacement temperatures of the individual flows, to the time intervals between successive flows, and to irregularities in the underlying topography. Each member constitutes a compound cooling unit or composite ash-flow sheet (Smith, 1960) that shows no complete cooling breaks; hence, each member is a time-stratigraphic unit that has a very short emplacement span.

Although the internal stratigraphy of the members is complex, systematic changes in physical and chemical properties permit correlation of units within each member with a high degree of confidence. The phenocryst mineralogy of the Bandelier Tuff is at first glance monotonously simple: major quartz, sanidine, and minor clinopyroxene, fayalite, and oxides. However, preliminary studies (Smith and Bailey, 1966) show that in the upper member, and probably also in the lower member, systematic chemical and mineralogical changes

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are recorded in pumices from successively emplaced flows. In the upper member, the phenocryst content changes from about 5 percent in the basal air-fall pumice and lower ash flows to more than 30 percent in upper units. In the uppermost unit, anorthoclase and hypersthene take the place of sanidine and fayalite, and there is a marked decrease in quartz content. This upper anorthoclase unit is unique in the Jemez Mountains and is critical to the correlation of the Bandelier Tuff inside and outside the Valles Caldera. It is this unit that dominates the outcrop on the structural dome within the caldera, whereas only a few remnants have survived erosion outside the caldera.

Intracaldera Rhyolites

The rhyolites within the Valles Caldera may be conveniently divided into three groups that are informally designated "early," "middle," and "late" on the basis of stratigraphy and structure (Smith and others, 1961).

The early rhyolite (S32) erupted on the caldera floor immediately after subsidence and rests conformably on the anorthoclase unit. It is overlain by sedimentary caldera fill, which accumulated locally to a depth of over 2000 feet mainly from erosion and slumping of the caldera walls. The middle rhyolite (S33), for the most part, overlies this fill and was erupted and contemporaneously faulted and tilted during uplift of the caldera floor — uplift that culminated in development of a large structural dome in the center of the caldera. The late rhyolite erupted in the moat that developed between this central structural dome and the caldera walls; it forms a discontinuous ring of lava domes and flows that are inferred to have been extruded from the major ring fracture that bounds the subsided, and subsequently uplifted, caldera block. The late rhyolite is not deformed or tilted; hence it postdates the central structural dome.

Field relations permit separation of the various units within the late rhyolite into four time-significant subgroups: Cerro del Medio, an accumulation of at least three thick rhyolite flows (designated I, II, and III, in order of decreasing age) is the oldest late rhyolite unit and is referred to as group A (*see* Pl. 1 for unit numbers of the various late rhyolite units).

The rhyolite domes and dome clusters of Cerro del Abrigo, Cerro Santa Rosa, Cerro San Luis, and Cerro Seco are, as a group, next youngest in age, but their relative ages are incompletely known. These domes will be referred to as group B. Cerro San Luis and Cerro Seco are single-vent units where they are exposed, whereas Cerro del Abrigo and Cerro Santa Rosa have younger satellitic domes, whose age relations with their larger neighbors are not completely known from field relations. The satellites are designated on the map (Pl. 1) and in the text by Roman numerals II and III. Roman numeral I in each case is reserved for the principal dome or oldest lava flow of a given cluster. The domes and flows of San Antonio Mountain, South Mountain, and probably Cerro La Jara constitute a third, still younger group (group C) of late rhyolite units, although their age relation to one another is not known on the basis of field relations alone. Petrologic and chemical

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progression suggests a temporal succession of the principal domes from Cerro del Medio westward to San Antonio Mountain, and, as will be discussed later in this paper, the K-Ar and paleomagnetic studies tend to confirm this general succession. Warm Springs dome, a small endogenous rhyolite dome that is now being exhumed from young lake beds and alluvium may belong to this subgroup, but its stratigraphic relationship with the other units is not certainly known.

The youngest rhyolite units in the Valles Caldera are the ash flows of Battleship Rock and the overlying obsidian flow of Banco Bonito, both of which apparently issued from the same vent.

Petrographically, the rhyolites of Cerro del Medio are almost without phenocrysts, but an occasional small sanidine crystal may be seen. The other late rhyolite units, discussed here, contain phenocrysts of quartz, sanidine, biotite, and oxides, and show a general increase in femic minerals and total phenocryst content westward from Cerro del Medio. Hornblende and plagioclase are common in the youngest, westernmost units (San Antonio Mountain, Battleship Rock, and Banco Bonito).

Intracaldera Hydrothermal Alteration

All rock units within the caldera, except for most of the late rhyolite units, show the effect of hydrothermal or other aqueous alteration. Soon after subsidence, the Valles Caldera was occupied by a lake that persisted, at least locally, until very late in the late rhyolite stage. Most of the caldera fill, the early rhyolite, and the middle rhyolite were probably deposited in water. The subsided floor of the caldera, later uplifted to form a structural dome, was locally buried by more than 2000 feet of caldera fill and has been the site of considerable hydrothermal activity, some of which still persists. The rocks of the western and northern quadrants of the structural dome (Pl. 1) are in general more altered than the eastern and southern quadrants.

Away from the obviously argillized hydrothermal aureoles, the alteration is predominantly pervasive silicification and oxidation, often so subtle as to almost escape notice. Locally, seams of chalcedonic silica or coarser quartz are common, as are opaline deposits in cavities and fractures. Of particular significance as evidence for a pervasive, but incomplete, alteration within the structurally domed area is the widespread conversion of tridymite and cristobalite to quartz (Smith, 1960, p. 152) in the Bandelier Tuff units. Whereas outside the caldera, groundmass quartz is virtually unknown in the Bandelier Tuff, inside it is difficult to find specimens without some groundmass quartz. Spherulites and other devitrified zones in the early and middle rhyolite also show that conversion of cristobalite to other forms of silica has taken place. Although small local areas occur that seem to have escaped alteration of any kind, it is probable that none of these areas has been entirely free of the solutions that caused pervasive alteration elsewhere.

The late rhyolite units are remarkably free of alteration of any kind except partial hydration of their glassy facies. The one known exception is the Warm

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Springs dome, which shows considerable veining with quartz, chalcedony, and opal in parts of its outcrop area. Individual specimens appear fresh, but again these specimens have probably been subjected to warm solutions from the associated hot spring at a time when both were submerged in the caldera lake. A wave-cut terrace on top of Warm Springs dome attests to its subaqueous history.

PALEOMAGNETIC STUDIES

Methods

The experimental factors that control paleomagnetic reliability have been discussed in earlier papers (Cox and others, 1963; Doell and others, 1966). Briefly, the only practical criterion is to ascertain that the primary stable component of the natural remanent magnetization (NRM) has been clearly identified and measured. This requires the collection of multiple samples from the igneous unit in question and the application of partial demagnetization techniques to remove the less stable secondary components of remanent magnetization that may have been imparted by the effects of lightning (Cox, 1961; Graham, 1961) or by long exposure in the earth's magnetic field (Rimbert, 1956). It is also necessary to show or imply that the stable primary magnetization is thermoremanent magnetization (TRM) acquired at the time the igneous unit cooled.

At each sampling site (Pl. 1) from 5 to 12 oriented cores were collected using equipment and techniques described by Doell and Cox (1967d). In general, the cores were collected over lateral distances of tens of meters so as to obtain a representative direction of remanent magnetization for each igneous body (sampling extents and number of cores collected from each unit are given in Table 1). Spinner magnetometers (Doell and Cox, 1967b) were used to measure remanent magnetization, and, for the most part, partial alternating-field demagnetization (Doell and Cox, 1967a) was used for testing paleomagnetic reliability and for magnetic cleaning.

Because of the small percentage of ferromagnetic minerals in the rocks studied, compared with basalts and andesites, the NRM intensities are generally very low. This made it difficult to detect in the field (Doell and Cox, 1967c) and to avoid those exposures that had been remagnetized by lightning. Consequently, we fortuitously collected several cores that had been affected by lightning-induced magnetic fields greater than the maximum alternating-field strength that was available in the demagnetization equipment (2400 oersteds peak value). Therefore, it was not possible to identify and measure the original TRM in all samples. Hydrothermal activity in parts of the Valles Caldera also has affected the reliability of some of the magnetic measurements. Fortunately, however, it was possible to detect many of these adverse effects by laboratory experiments and to exclude such data from the compilation of paleomagnetic results (Table 1).

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Discussion of Paleomagnetic Results

It is convenient to discuss the paleomagnetic studies in reverse stratigraphic order. The two youngest units studied, the obsidian flow of Banco Bonito (S47) and the welded tuff of Battleship Rock (S46), are both normally magnetized. Less stable components of the NRM were easily removed by alternating-field demagnetization at 400 oersteds peak value, and demagnetization curves made on representative samples are typical of TRM (Doell and Cox, 1963). It was also possible to collect samples from a baked zone beneath the obsidian flow of Banco Bonito and from inclusions of older rock in the welded tuff of Battleship Rock; these samples possess NRM directions that closely parallel samples of the baking units.

Most of the cores that were collected from the units of group C contained relatively large amounts of unstable remanent magnetization. Most of this unstable magnetization was removed by alternating-field demagnetization at the peak-field strengths given in Table 1. Results of the partial alternating-field demagnetization experiments that were performed on the samples from San Antonio Mountain II (S45) are typical of results for all four domes. The very scattered directions of NRM for this dome are shown in Figure 2(A). Three samples, numbers 4D068, 4D071, and 4D072, were chosen for successive partial alternating-field demagnetization at peak-field strengths of 12.5, 25, 50, 100, 200, 400, 600, and 800 oersteds, and the results of these experiments are given in Figure 3. Two components of remanent magnetization are clearly present in these three samples. The component with coercive forces generally greater than 400 oersteds and with a similar direction in all three samples was taken to be the stable original TRM. A peak field of 400 oersteds was then chosen for magnetic cleaning of the remaining five samples, and the resulting directions are shown in Figure 2(B). Sample number 4D067 did not respond to this treatment and was further demagnetized up to peak field strengths of 2000 oersteds (Fig. 3A). This sample possesses a stable component of magnetization parallel to that in the other samples, but, in addition, it has an unusually large secondary component that was probably induced by lightning.

Partial alternating-field demagnetization, although very effective in removing viscous and lightning-induced components of remanent magnetization, also adds an anomalous magnetization (Doell and Cox, 1967a), the intensity of which increases with the peak alternating-field strength that is employed. The amount of anomalous magnetization that is given to a rock, compared with the amount of remanent magnetization that remains after treatment, varies considerably from sample to sample, but increases rapidly at higher alternating-field strengths. For this reason, magnetic cleaning of all samples from San Antonio Mountain II at higher peak-field strengths was not carried out, and sample 4D067 was excluded from the analyses given in Table 1. Although there are better methods for choosing the most effective field strength for magnetic cleaning (Irving and others, 1961), they consume more

TABLE 1. PALEOMAGNETIC DATA FOR ROCKS OF THE VALLES CALDERA, NEW MEXICO

Name and Unit No.		Site location lat	Site location long	Lateral distance* (meters)	N†	N'‡	$\bar{H}$§ (oe)	RM direction|| Decl	RM direction|| Incl	R#	ξ_c**	k_c††	$\propto_{95}$‡‡	VGP Position§§ lat	VGP Position§§ long
Banco Bonito	S47	35.88°	253.37°	1000	8	8	400	359.0°	41.3°	7.726	0.282	26	11.2	77.8°	77.6°
Battleship Rock	S46	35.86	253.36	5000	9	9	400	358.3	43.2	8.975	.079	317	2.9	79.2	81.6
San Antonio II	S45	35.94	253.40	40	8	7	400	7.6	55.2	6.875	.204	48	8.8	83.8	343.6
South Mountain	S44	35.82	253.46	7000	11	10	600	357.0	56.5	9.621	.292	24	10.1	87.3	191.4
La Jara	S43	35.86	253.50	320	8	7	200	354.0	51.9	6.980	.081	306	3.5	84.0	131.5
San Antonio I	S42	35.90	253.36	3000	8	7	200	8.8	59.8	6.926	.157	81	6.7	81.6	305.9
San Luis	S41	35.94	253.45	1400	12	9	NRM	195.1	−56.8	8.987	.057	609	2.1	−77.8	152.2
Santa Rosa II	S40	35.96	253.50	30	8	7	NRM	194.8	−55.1	6.954	.124	129	5.3	−78.0	160.7
Seco	S39	35.95	253.44	30	8	2	400	165.5	−70.6	1.998	.057	622	10.0	−68.6	50.1
Santa Rosa I	S38	35.95	253.50	30	8	7	200	99.4	−67.4	6.980	.081	305	3.5	−32.4	25.1
Abrigo III	S37	35.96	253.51	70	7	7	200	23.7	50.8	6.903	.180	62	7.7	69.9	349.3
Warm Springs Dome	S36	35.97	253.44	60	7	....	....	....	....	....	....	....	....	....	....
Del Medio I	S35	35.91	253.52	20	7	7	200	213.6	−44.6	6.954	.123	132	5.3	−59.8	172.6
Del Medio II	S34	35.92	253.54	10	8	8	200	178.9	−48.5	7.911	.160	79	6.3	−83.4	262.1
Middle rhyolite	S33	35.92	253.39	8000	36	....	....	....	....	....	....	....	....	....	....
Early rhyolite	S32	35.88	253.40	20	5	....	....	....	....	....	....	....	....	....	....

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TABLE 1. (CONTINUED)

Name and Unit No.		Site location lat	long	Lateral distance* (meters)	N†	N'‡	H̃§ (oe)	RM direction|| Decl	Incl	R#	ξ_c**	k_c††	$\propto_{95}$‡‡	VGP Position§§ lat	long
Bandelier Tuff															
Upper member															
"Anorthoclase" unit		35.83°	253.66°	64	7	7	200	174.2	−35.5	6.989	.060	560	2.6	−73.0	272.6
Site 5		35.77	253.26	40	6	4	400	174.8	−34.2	3.997	.042	1116	2.8	−72.4	270.0
Site 4		35.77	253.26	30	6	6	400	176.6	−32.9	5.991	.061	542	2.9	−71.9	263.7
Site 3		35.77	253.26	20	6	6	400	173.4	−34.8	5.958	.130	119	6.2	−72.4	274.7
Site 2		35.77	253.26	20	6	6	400	169.3	−26.6	5.932	.165	74	7.9	−66.3	279.7
Site 1		35.83	253.63	60	7	7	200	173.4	−37.7	6.968	.104	185	4.4	−74.3	275.7
All Upper member	S31	35.77	253.26	38,000	38	36	....	173.6	−33.8	35.748	.120	139	2.0	−71.8	273.1
Lower member															
Site 6		35.84	253.35	30	6	5	200	172.5	−44.5	4.981	.098	207	5.6	−78.4	289.1
Site 5		35.84	253.35	20	6	6	200	170.8	−48.6	5.986	.075	355	3.6	−80.1	306.9
Site 4		35.93	253.29	30	6	6	200	174.8	−44.5	5.982	.083	285	4.0	−79.2	279.2
Site 3		35.93	253.29	20	6	5	200	171.3	−44.7	4.985	.086	268	5.0	−77.9	293.3
Site 2		35.93	253.29	10	6	6	200	176.1	−47.1	5.972	.105	182	6.0	−81.7	277.9
Site 1		35.93	253.29	30	6	6	200	177.8	−45.2	5.987	.071	398	3.4	−80.6	265.5
All Lower member	S30	35.93	253.29	10,500	36	34	200	174.0	−45.9	33.865	.091	244	1.6	−80.0	285.4
Sawyer dome	S29	35.85	253.56	20	5	5	200	12.2	44.4	4.976	.109	169	5.9	75.7	23.3

*Lateral Distance = Maximum lateral distance between cores in meters
†N = Number of sample cores collected
‡N′ = Number of sample cores used in analysis
§H = Peak strength of alternating field used for magnetic cleaning
||RM direction = Declination and inclination of remanent magnetization
#R = Vector sum of N′ unit vectors
**ξ_c = Intraflow angular standard deviation factor (Cox and Doell, 1964)
††k_c = Intraflow precision factor = (N-1)/(N-R) (Fisher, 1953)
‡‡$\propto_{95}$ = 95% confidence cone about average direction (Fisher, 1953)
§§VGP Position = Latitude and longitude of virtual geomagnetic pole (Cox and Doell, 1960)

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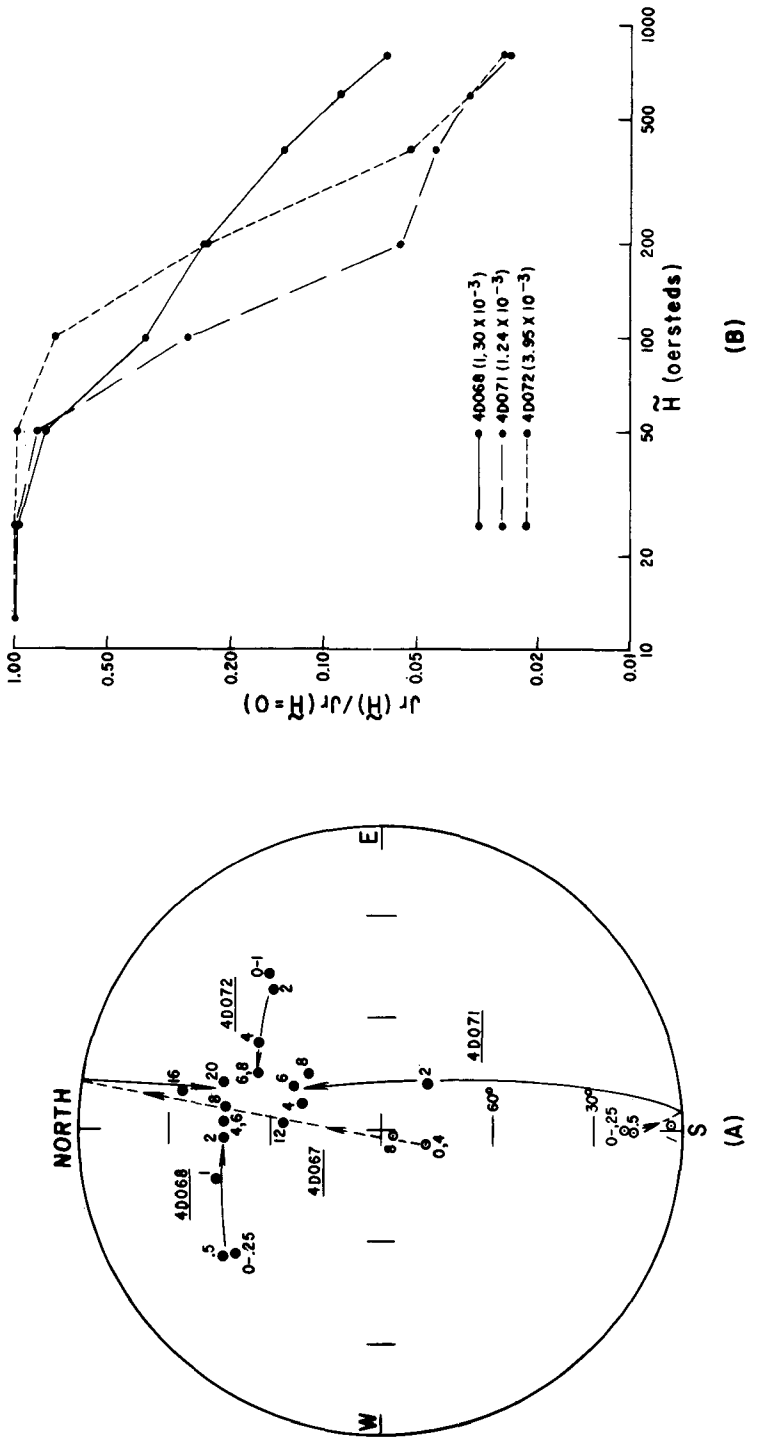


Figure 3. Changes in remanent magnetization for three selected samples from San Antonio Mountain II (S45) during progressive alternating field demagnetization. (A) Directional changes — plotting notation as for Figure 2. Numbers adjacent to points indicate peak alternating-field strengths in hundreds of oersteds. (B) Intensity of remanent magnetization, $J_r(\tilde{H})$, relative to the intensity of NRM, $J_r(H=0)$ (ordinate), after treatment in an alternating field of strength $\tilde{H}$ (absissa). NRM intensities in c.g.s. e.m.u./c.c. shown in parentheses.

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time, and the experiments that are performed here identify and define the direction of the original TRM satisfactorily for the present purposes.

Similar demagnetization experiments and reasoning led to the data in Table 1 for the other three domes in this group. All have well-defined normal directions of stable remanent magnetization.

The group B units sampled were the six rhyolites of Cerro San Luis (S41), Cerro Santa Rosa II (S40), Cerro Seco (S39), Cerro Santa Rosa I (S38), Cerro del Abrigo III (S37), and Warm Springs dome (S36). Except for Cerro Seco and Warm Springs dome, there is relatively little scatter of the NRM directions of these domes, and partial-demagnetization treatments similar to those described above were used to identify and define the stable remanent magnetization.

All eight samples from Cerro Seco, which possessed quite scattered NRM directions, were partially demagnetized at peak-field strengths of 12.5, 25, 50, 100, 200, 400, 600, and 800 oersteds (Fig. 4). One, number 4D074, was further demagnetized at peak values of 1200 and 1600 oersteds. For only two samples, numbers 4D078 and 4D080, was it possible to remove the secondary components of magnetization and identify the direction of the stable TRM. However, the directions in the other six samples all moved towards this same stable reversed direction as more and more of the unstable components were moved.

The NRM intensity in the seven samples that were collected from Warm Springs dome ($\approx 5 \times 10^{-5}$ gauss) is considerably weaker than the intensity observed in samples from the other domes of this group ($> 5 \times 10^{-4}$ gauss). Demagnetization experiments with peak field strengths above 400 oersteds were not successful, nor was it possible to identify a stable component of magnetization in any sample. It is also clear that this sampling site has been involved in hydrothermal activity, and although obvious alteration was avoided in sampling, the original magnetization may have been destroyed or changed by chemical alteration.

Three rhyolites of this group, Cerro Santa Rosa II, Cerro San Luis, and Cerro Seco have reversed directions of remanent magnetization; one (Cerro Santa Rosa I) has a direction that is best described as intermediate; and one (Cerro del Abrigo III) has a well-defined normal direction of remanent magnetization. Because the ancient geomagnetic field directions that are determined from this group are the primary data for defining the Jaramillo normal-polarity event (Doell and Dalrymple, 1966), several additional studies were performed on these rocks to test for a possible self-reversal of remanent magnetization (Uyeda, 1958; Verhoogen, 1962) in one or more of these domes. Because ferromagnetic minerals possessing such a self-reversal property should be different from those that do not, these additional experiments involved measurements of intrinsic magnetic properties.

The mode of decay of remanent magnetization with progressive partial alternating-field demagnetization in the different domes is shown in Figure 5. Except for the sample from Cerro Seco, all samples had NRM directions

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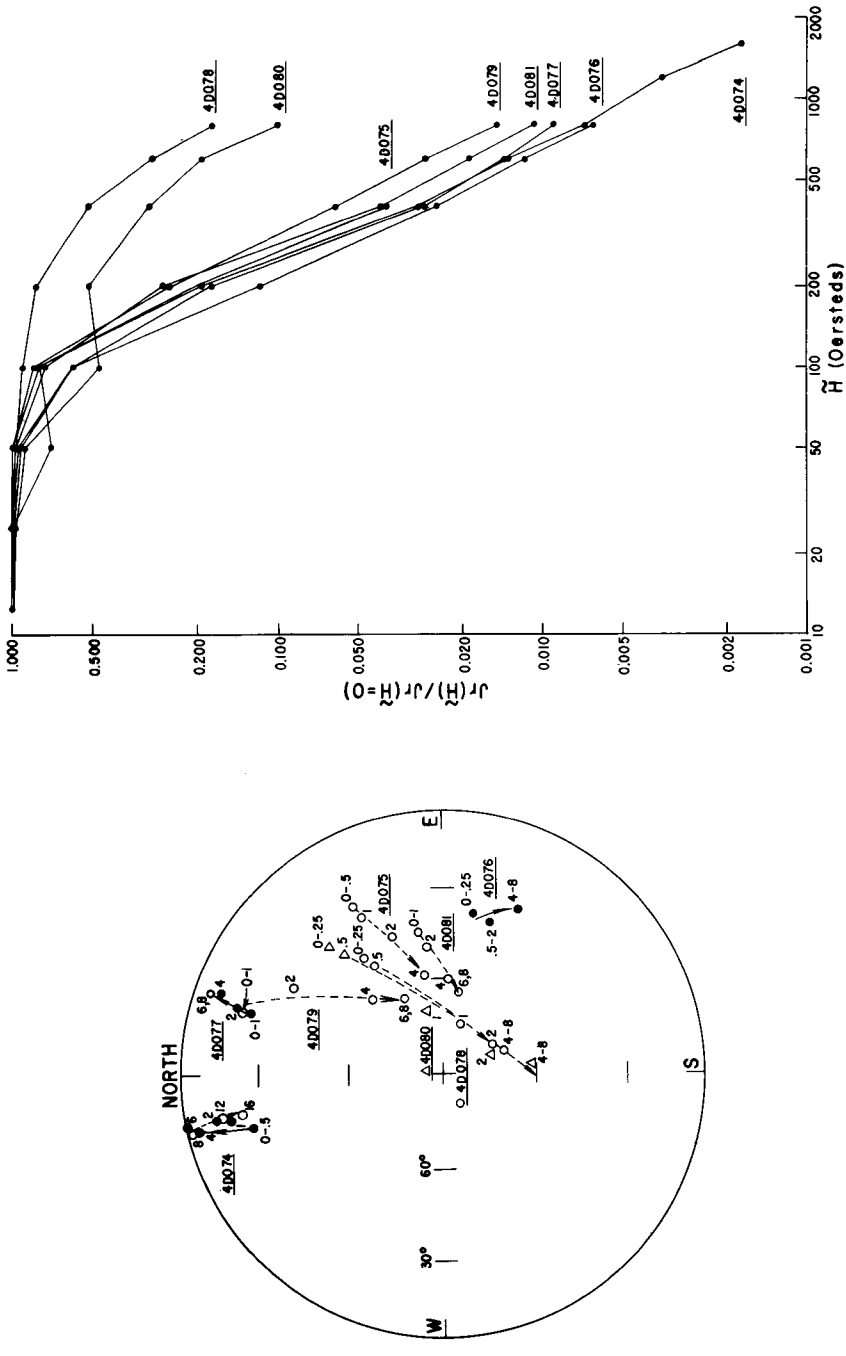


Figure 4. Changes in the direction (A) and intensity (B) of remanent magnetization for individual samples from Cerro Seco (S39) during progressive alternating-field demagnetization. Conventions as in Figure 3.

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near the average direction of stable magnetization for the dome in question, and these directions did not change throughout the demagnetization experiments. The sample from Cerro Seco (number 4D078) had a relatively small component of unstable magnetization that was removed after partial demagnetization at 200 oersteds peak value (*see also* Fig. 4). These curves deter-

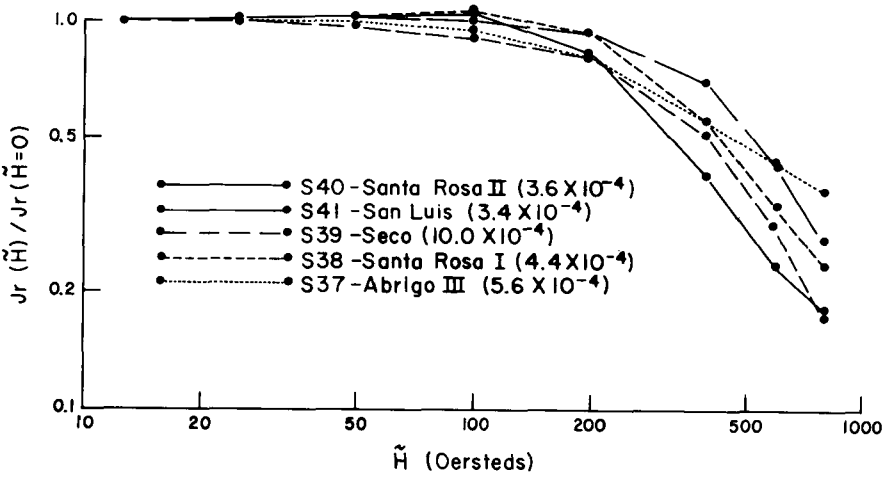


Figure 5. Changes in the intensity of remanent magnetization with progressive alternating-field demagnetization for selected samples from five rhyolite domes. Numbers in parentheses are NRM intensities in c.g.s. e.m.u./c.c. Other conventions as in Figure 3B.

mined from samples that possess minimal secondary magnetization are remarkably similar, as is the intensity of NRM in the different samples; thus, there is no evidence of differences in these properties that might be related to the different directions of stable magnetization.

Measurements of saturation magnetization as a function of temperature between room temperature and 670° C made on specimens from two different sample cores that were collected from Cerro San Luis, Cerro Santa Rosa I, and Cerro del Abrigo III, and from one sample core that was collected from Cerro Santa Rosa II and Cerro Seco are shown in Figure 6. They were obtained with the apparatus described by Doell and Cox (1967e) using a vacuum of $< 10^{-4}$ torr and an applied magnetic field of 7300 oersteds. The most obvious features of these curves are that all samples underwent chemical change during the experiments and that there is about as much difference between samples from the same dome as between samples from different domes. The curves yield no evidence of a mineralogic difference that is associated with different magnetization directions.

The mode of thermal decay of NRM in specimens cut from the same samples used for the saturation magnetization measurements is shown in Figure 7. These data were obtained by heating the samples in a vacuum of

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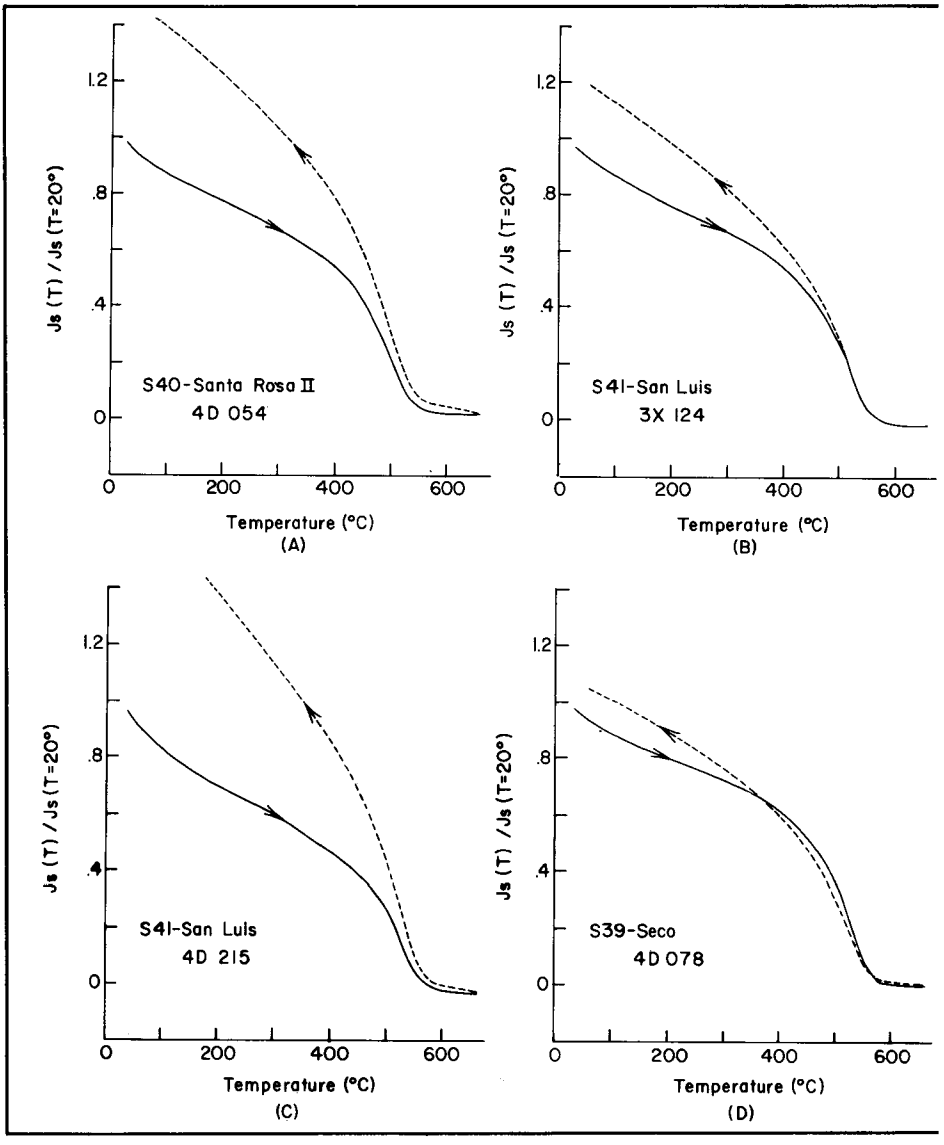
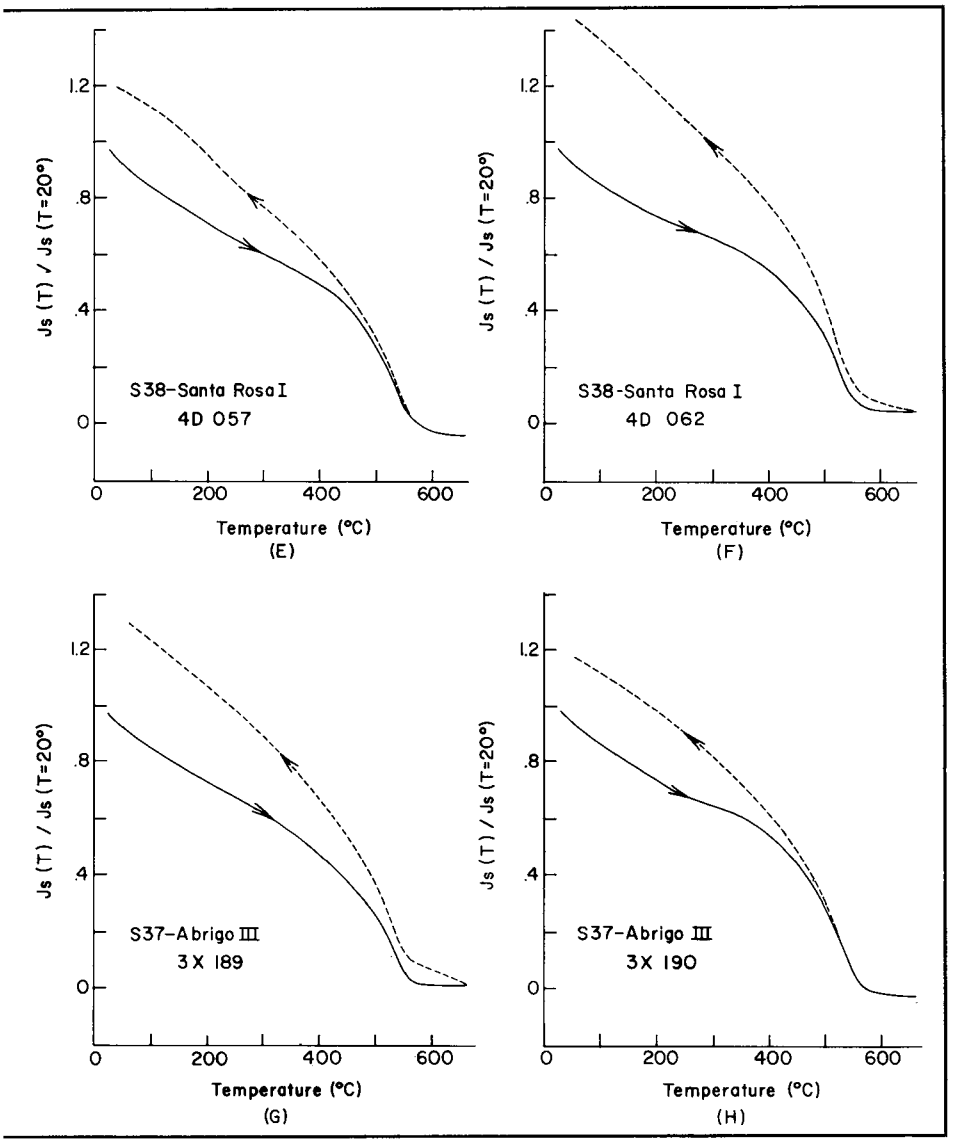


Figure 6. Saturation magnetization versus temperature for the samples listed. Ordinate is magnetization, J_S, at temperature T, relative to magnetization at room

$< 10^{-4}$ torr to the temperatures that are indicated and cooling the samples in a magnetic field-free space, after which the remaining remanent magnetization was measured (Cox and others, 1963). All samples were also partially demagnetized in an alternating field of 100-oersteds peak value before they were heated to remove less stable components of magnetization. There were

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temperature (T=20° C). Solid line is heating curve, dashed line, cooling curve. Measurements were made in a magnetic field of 7300 oersteds and a pressure of $< 10^{-4}$ torr.

two components of remanent magnetization in each sample, one with a Curie temperature near 550° C and the other with a Curie temperature between 625 and 700° C. Moreover, these two components, within a given sample, were aligned in the same direction. Figure 8 shows directions of remanent magnetization throughout these experiments for three representative specimens.

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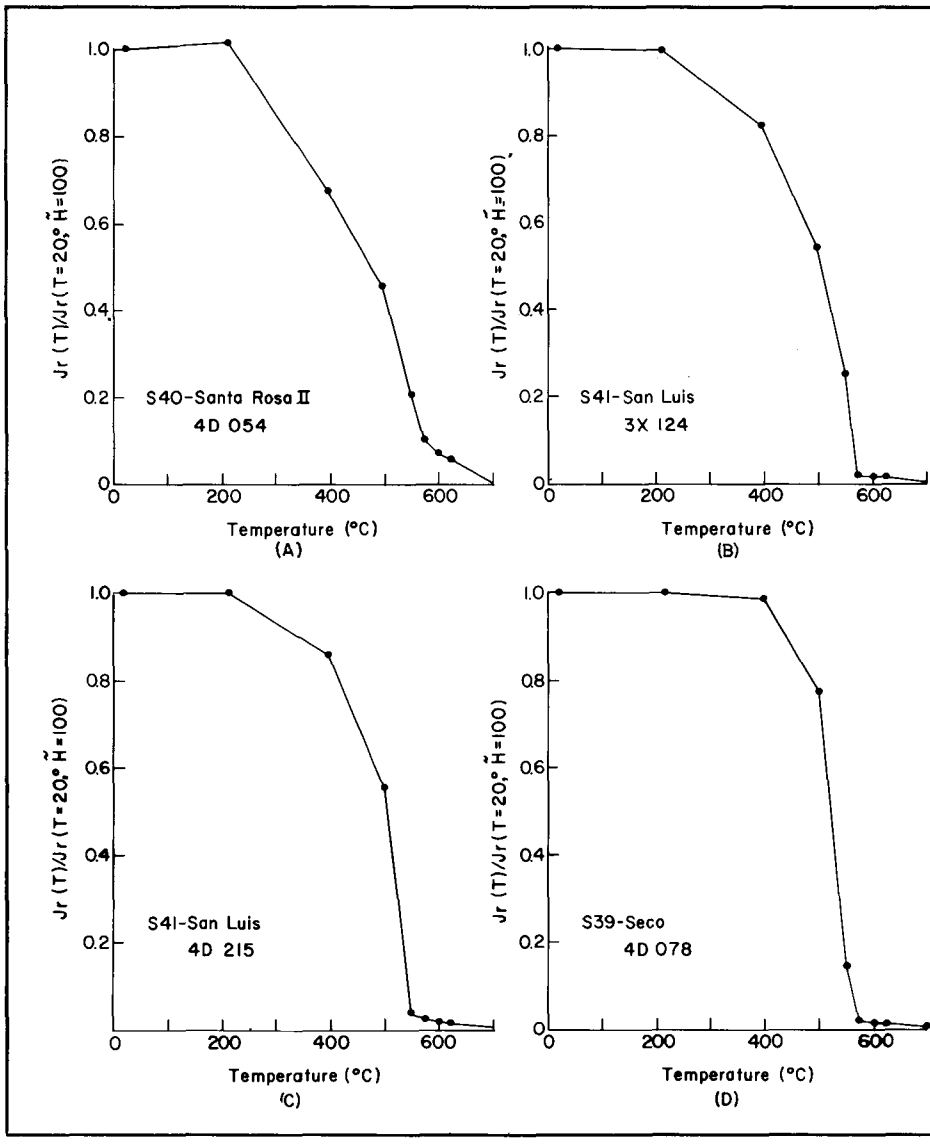
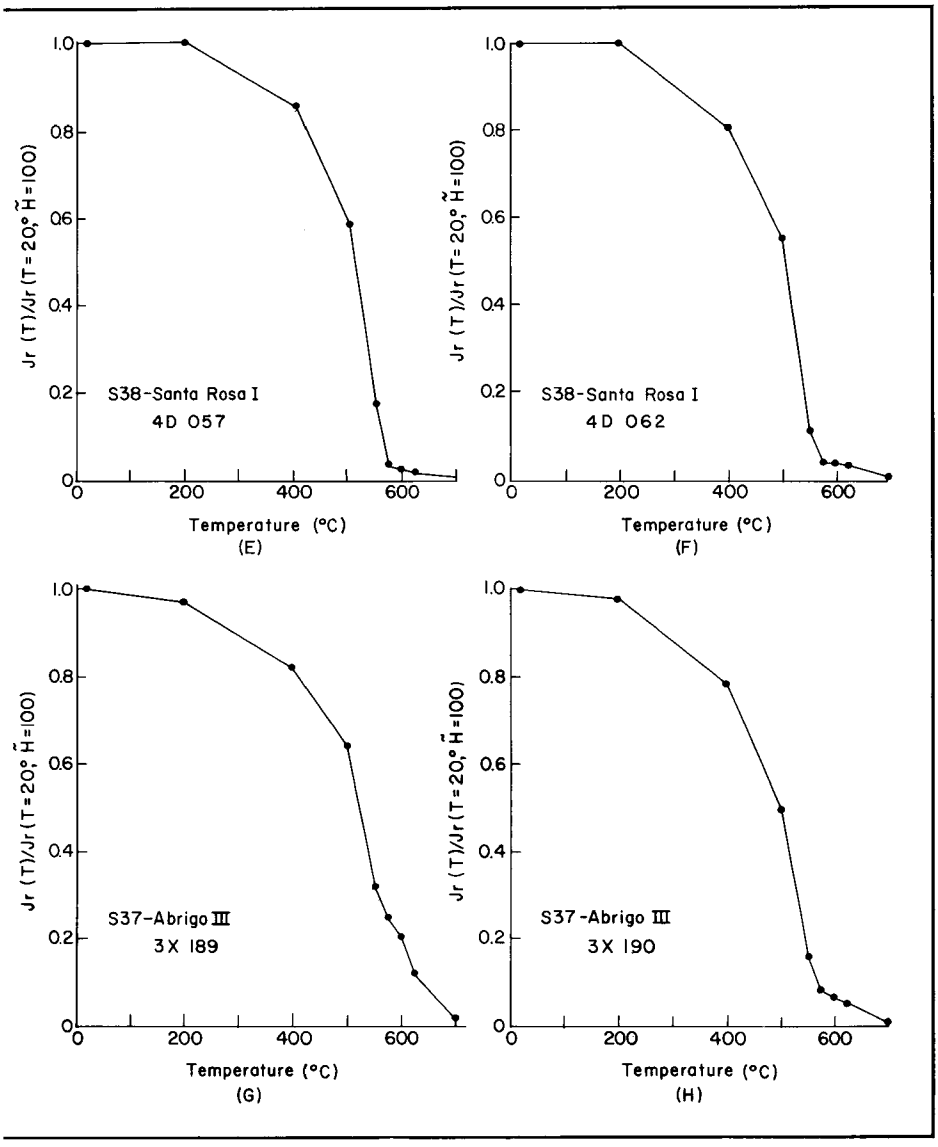


Figure 7. Decay of NRM with temperature for the samples listed. Ordinate is remanent magnetization remaining after heating the samples to the temperatures indicated and cooling to room temperature in a field-free region (relative to RM before heating).

If these two components were present at the time the rock originally cooled and acquired its NRM, then there would be a strong case against a self-reversal of the NRM (Cox and others, 1963). However, the saturation magnetization curves indicate that a chemical change takes place during heating, and it is possible that one of the components seen in Figure 7 was not present

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The samples were heated and cooled in a vacuum of $< 10^{-4}$ torr and were partially demagnetized in an alternating field of 100-oersteds peak value before the first heat treatment.

in the natural rock. If this were so, though, it is difficult to imagine why the new component would always have the same direction of magnetization as the original component. It is likely that the two components are original, but there are other possibilities that cannot be excluded on the basis of the present data, and additional experiments are planned. For the present, it may be

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noted that all five of the rhyolites exhibit these same characteristics and that there are no indications in the curves of Figure 7 for the existence of two different remanent magnetizations with opposing directions. All samples were also cooled from 700° C in a known field of about 0.5 oersted and acquired TRM in the direction of this field, but because of the chemical changes mentioned above, these results also are not entirely conclusive.

On the basis of these observations, especially the intermediate direction of stable magnetization in Cerro Santa Rosa I, the existence of a self-reversal of the natural remanent magnetization is very unlikely in any of these domes. We therefore feel that the stable remanent magnetization is original TRM that was acquired in the direction of the geomagnetic field that existed at the time the rocks formed.

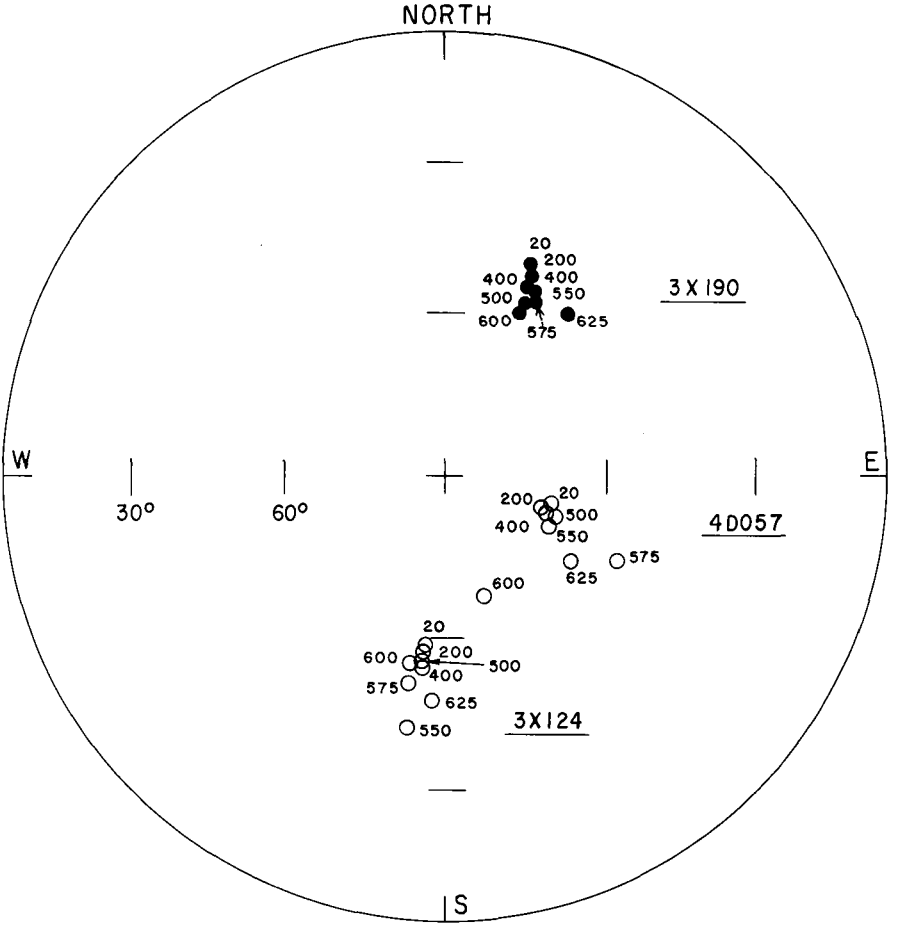


Figure 8. Directions of remanent magnetization during mode of thermal decay experiments for three of the samples shown in Figure 7. Numbers adjacent to data points indicate temperature in degrees C. Plotting conventions as in Figure 2.

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The rhyolites of group A, Cerro del Medio I (S35) and Cerro del Medio II (S34), were relatively unaffected by unstable components of remanent magnetization. Demagnetization curves for selected samples and magnetic cleaning at 200-oersteds peak value clearly identified a stable reverse remanent magnetization in both domes.

Exposures of the early rhyolite (S32) that are suitable for paleomagnetic studies are limited, and samples were collected from only one site (Pl. 1). However, the middle rhyolite (S33) is more widespread, and because of its importance to the geological studies, a total of 36 cores was collected from the three sites that are indicated in Plate 1. The NRM directions of the early rhyolite are badly scattered, and it was not possible to recognize a stable component of remanent magnetization in any sample; all were demagnetized in progressive steps to a peak field value of 800 oersteds. NRM directions of the middle rhyolite also are badly scattered. Progressive demagnetization experiments were performed on 13 representative samples, but only in one or possibly two samples was there an indication of a stable component of magnetization. Moreover, unlike the behavior of the scattered directions in Cerro Seco, which all moved towards a stable reversed direction during progressive demagnetization, the changes in direction of the 13 middle rhyolite samples were quite random during these experiments. For these reasons, and the fact that there is evidnce of hydrothermal alteration, the results from these two units are unreliable and, unfortunately, cannot be used in the present study.

The two members of the Bandelier Tuff (S31) (S30), are made up of several "levels" with varying characteristics. The geologic studies indicate a significant time interval between the two members, but not between different levels within a given member. To further test these findings, we collected samples from several stratigraphic levels within each member (Pl. 1; Fig. 9). Secular variation of the geomagnetic field commonly causes changes in direction of several tens of degrees at a given locality within a period of time of a few hundred to a few thousand years. Thus, significantly different average directions of remanent magnetization from one stratigraphic level to another, or between one member and the other, would indictae time intervals between subsequent depositions greater than this length of time. Identical directions offer permissive, but not conclusive, evidence that the depositions closely follow one another.

Although components of unstable magnetization were present in some samples of the Bandelier Tuff, little difficulty was experienced in identifying and defining the stable direction of remanent magnetization. All samples are reversely magnetized.

Ninety-five percent confidence intervals of the magnetic directions for the lower and upper members are plotted in Figure 10, and it is clear that they strongly support the geologic conclusions concerning rapid accumulation of the two individual members with a long interval between them.

The five samples collected from Sawyer dome (S29) possessed only a

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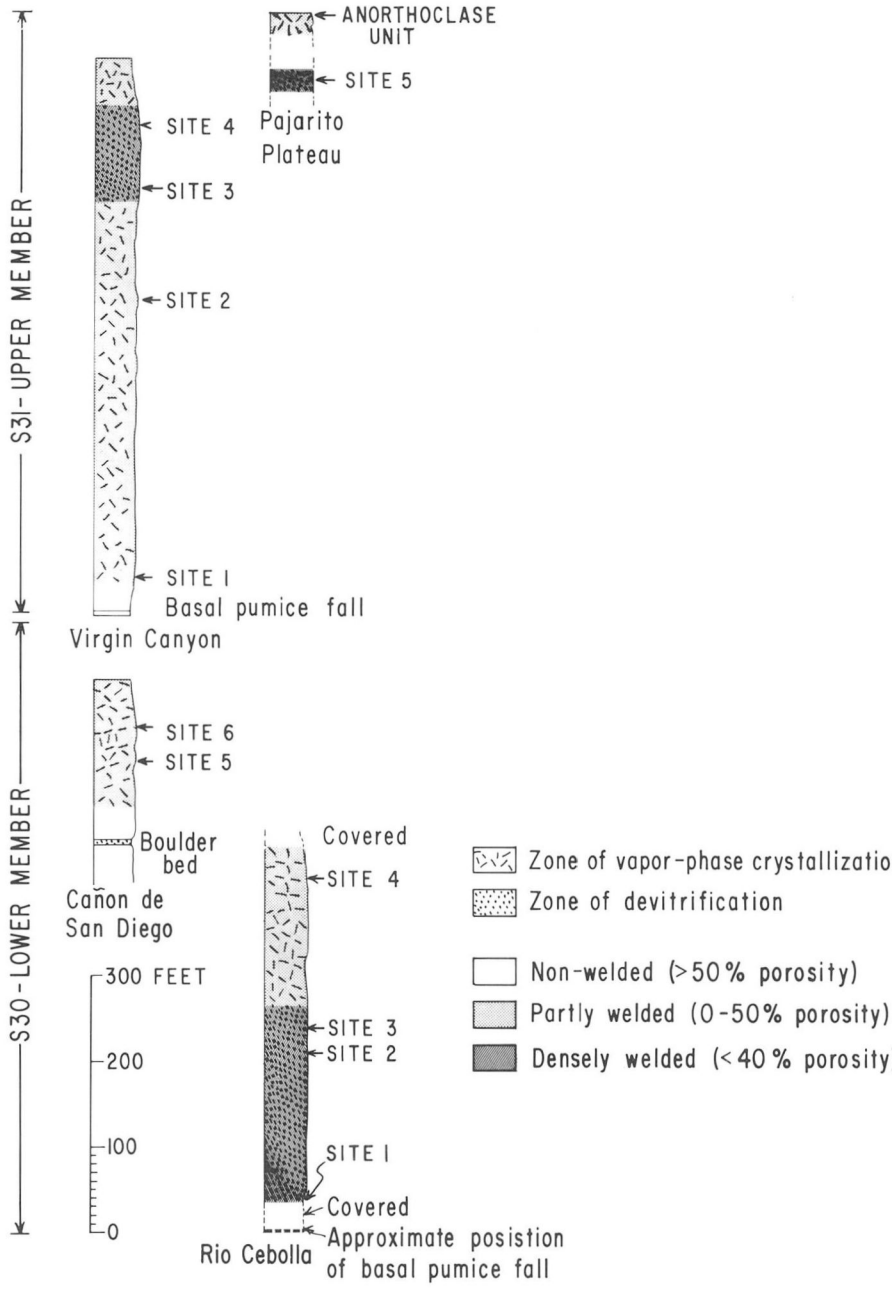


Figure 9. Composite stratigraphic column of the Bandelier Tuff and stratigraphic position of samples.

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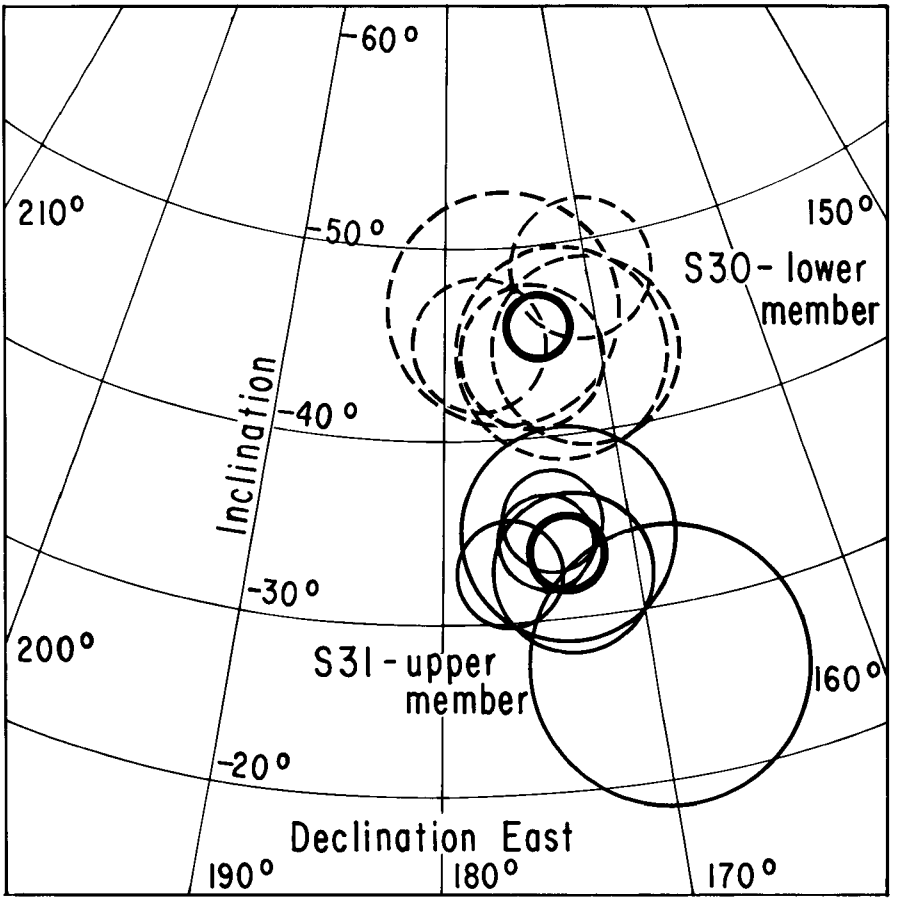


Figure 10. Ninety-five per cent confidence limits for the directions of magnetization of different levels within the lower member (S30) and upper member (S31) of the Bandelier Tuff (light circles). Heavy circles are for all samples within each member. Upper hemisphere of a portion of the Lambert equal-area projection.

small amount of unstable remanent magnetization. A stable normal component was identified and defined by progressive demagnetization and magnetic cleaning at 200-oersteds peak value (Table 1).

Because the rhyolite domes, from which the samples for this study were collected, often exhibit flow banding, the possibility of a magnetic anisotropy, which could influence the direction of stable remanent magnetization (Uyeda and others, 1963), must be considered. To test this idea, "high coercivity" magnetic anisotropy measurements were made on samples from nine different units (Battleship Rock, San Antonio I, San Luis, Santa Rosa I, Abrigo III, Del Medio II, anorthoclase unit, upper member of the Bandelier Tuff site 3, and lower member of the Bandelier Tuff site 4) using a modification of the method described by Cox and Doell (1967). Briefly, the method consists of imparting a "saturation" isothermal remanent magnetization (IRM) to the

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sample in a known direction. The sample is then partially demagnetized at a relatively high peak alternating-field value, and the remaining magnetization is measured. After complete demagnetization of the sample, the sequence is repeated, but with the IRM induced in a different direction. If anisotropy is present, the RM measurements will have a direction that is displaced from that of the inducing field, and its intensity will be different in the different directions. Because TRM also resides in relatively high coercive-force regions, an anisotropy that might cause original TRM to be acquired at an angle to the ambient magnetic field should also cause angular deviations in the present experiments.

For rocks of the Valles Caldera, an inducing field of 5900 oersteds was employed, followed by partial alternating-field demagnetization with a peak-field strength of 800 oersteds. IRM was induced in each sample in three different directions 45° from each other; one direction was along that of the stable remanent magnetization. The average difference between the measured RM (after demagnetization) and the inducing field was 4.7°; the maximum difference was 9.5°; and the deviations were unsystematic. Thus, there is no evidence of anisotropy in these rocks. Although the report of Cox and Doell (1967) indicated a "noise" level of about 2° for this procedure, the new device allowing IRM to be imparted in any direction is less accurate than the previous device, so that part of the larger deviations in the present study are due to differences in precision.

Except for the early and middle rhyolites, which did not yield meaningful magnetic data, no unit studied has been involved in tectonic movements, and no tilt corrections have been applied to the data in Table 1.

POTASSIUM-ARGON DATING

Analytical Techniques

The materials used for the potassium-argon determinations were hand specimens from an outcrop that was encompassed by the paleomagnetic samples. Mineral separates of feldspar or biotite were used for all analyses except those on Del Medio I (S35) and II (S34), in which the determinations were on obsidian (sample numbers 4D005 and 4D004, respectively). The sanidine from the basal ash falls of the lower and upper members of the Bandelier Tuff sheets was separated from hand-picked primary pumice fragments in order to minimize contamination by older xenolithic or xenocrystic material. For the other units, mineral separates were from representative specimens of the bulk rock. All of the feldspar concentrates were treated briefly with a cold 5 to 7 percent solution of hydrofluoric acid. This procedure leads to a purer mineral concentrate, helps to reduce the atmospheric argon contaminant, and has no known undesirable effects on the age determination (Evernden and Curtis, 1965).

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Aliquants of the mineral separates were taken for the potassium and argon analyses with a Jones-type microsplitter. For the two whole-rock obsidian samples, the analyses were done on small adjacent blocks cut from the hand specimen with a diamond saw.

A minimum of two potassium analyses was made on all samples. Except for the early rhyolite and the lower member of the Bandelier Tuff site 1, either argon analyses were duplicated or more than one sample from the same rock unit was analyzed.

Potassium was determined by flame photometry using a Baird KY-2 flame photometer with a lithium internal standard and natural gas and air for fuel. The chemical procedure is described by Shapiro and Brannock (1962).

Argon analyses were by isotope dilution. All samples were preheated at 200° C for 12 to 18 hours to reduce the atmospheric argon contamination. Samples were fused by induction heating and the evolved gas was cleaned using synthetic molecular sieve (Linde type 5A) to remove water, and hot CuO and Ti foil to eliminate reactive gases. A bulb tracer with an approximate tracer size of 3×10^{-10} mole Ar^{38} per release was used (Lanphere and Dalrymple, 1966). Mass analyses were made with a Reynolds-type mass spectrometer.

The analytical data are presented in Table 2. Potassium-argon ages (Table 3) were calculated using the constants $\lambda_\varepsilon = 0.585 \times 10^{-10}$ yr^{-1}, $\lambda_\beta = 4.72 \times 10^{-10}$ yr^{-1}, and $K^{40}/K_{total} = 1.19 \times 10^{-4}$ mole/mole. For the results of analyses on interlaboratory standard minerals done in our laboratory, see Lanphere and Dalrymple (1965).

Precision of the Age Determinations

It is desirable to have quantitative information about the statistical dispersion of age determinations for two reasons: (1) they serve as useful indicators of general reliability, and (2) they can be used to estimate the likelihood that two calculated ages are truly different and not simply due to random errors in the analytical procedure. Ideally, it is desirable to know how the "true" ages of two rock units compare, although this is not really possible in any direct way. It is possible, however, to make estimates of the precision (reproducibility) of the age analyses, and from this information to calculate the probability of a real difference in Ar^{40}_{rad}/K^{40} of the two rock or mineral samples.

Such precision estimates for the present data have been made in two different ways: (1) from the quality of the individual experiments, and (2) from a statistical analysis of replicate measurements. First, estimates of the standard deviation of precision, s, for the argon analyses (Table 2) were made using methods described by Cox and Dalrymple (1967). These values of s range between 3 and 10 percent and average about 4.5 percent. This is larger than the s value of 0.55 percent that was found by Lanphere and Dalrymple (1965) for 12 analyses on an 80-million-year-old muscovite, and 1.84 percent

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TABLE 2. ANALYTICAL DATA FOR POTASSIUM-ARGON AGE DETERMINATIONS ON ROCKS FROM THE VALLES CALDERA, NEW MEXICO

Sample		Argon				
No	Material	K_2O (weight percent)	Weight (gms)	$\frac{Ar^{40}\ rad}{Ar^{40}\ total} \times 100$	Ar^{40} rad/gm* (10^{-11} mole)	Ar^{40} rad/K^{40} (10^{-5})
4D069	sanidine	10.30	10.225	25.1	0.6575 ± 0.0394	2.539
		10.32	10.109	53.8	0.6650 ± 0.0200	
3X074	sanidine	10.08	10.100	73.8	0.7341 ± 0.0220	2.892
		10.13	10.075	39.3	0.7426 ± 0.0297	
3X203	sanidine	10.20	10.755	48.9	0.7327 ± 0.0220	2.935
		10.10	10.304	43.4	0.7722 ± 0.0270	
3X062	sanidine	10.17	10.396	36.3	0.7872 ± 0.0394	3.132
		10.24	10.904	59.0	0.8277 ± 0.0248	
4D003	sanidine	8.47	10.495	79.7	0.8663 ± 0.0303	4.134
		8.54	10.265	63.2	0.9101 ± 0.0364	
4D001	sanidine	8.55	10.827	52.7	0.9130 ± 0.0274	4.247
		8.57	10.505	60.8	0.9237 ± 0.0277	
3X122	sanidine	8.67	7.043	71.3	1.054 ± 0.095	4.185
		8.67				
4D214	sanidine	8.64	9.426	42.4	0.8100 ± 0.0405	4.048
		8.63	11.266	68.3	0.8740 ± 0.0350	
			10.834	63.8	0.9179 ± 0.0367	
			11.244	76.7	0.9296 ± 0.0372	
4D002	sanidine	8.91	10.386	79.4	1.154 ± 0.058	5.174
		9.00	10.303	56.7	1.187 ± 0.047	
3X189	sanidine	9.17	10.933	77.5	1.173 ± 0.035	5.186
		9.15	4.105	58.8	1.227 ± 0.037	
3X117†	sanidine	7.00	8.311	76.8	1.219 ± 0.043	7.111
		7.09	8.256	31.0	1.312 ± 0.079	
4D005	obsidian	4.61	5.734	23.6	0.7053 ± 0.0494	6.074
		4.61	5.541	22.6	0.7087 ± 0.0496	
		4.61				
		4.61				
4D004	obsidian	4.64	5.738	36.7	0.7979 ± 0.0319	6.718
		4.64	6.691	39.7	0.7669 ± 0.0307	
		4.55				
		4.63				
4D207†	biotite	7.84	6.949	35.2	1.977 ± 0.079	9.991
		7.83				
4D013†	sanidine	9.06	5.326	48.1	1.719 ± 0.086	7.487
		9.12				
	biotite	7.43	5.401	32.6	1.770 ± 0.124	9.360
		7.54				

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TABLE 2 (CONTINUED)

Sample			Argon			
No	Material	K_2O (weight percent)	Weight (gms)	$\frac{Ar^{40}_{rad}}{Ar^{40}_{total}} \times 100$	Ar^{40}_{rad}/gm* (10^{-11} mole)	Ar^{40}_{rad}/K^{40} (10^{-5})
3X114†	plagioclase	1.31 1.31	4.904	37.4	0.6295 ± 0.0630	19.03
3X110†	sanidine	7.09 7.09	2.614	70.3	1.308 ± 0.118	7.320
3X241	anorthoclase	5.24 5.34	6.224 7.276	57.0 46.2	0.8796 ± 0.0440 0.9812 ± 0.0491	6.963
5A057†	anorthoclase	5.62 5.67	5.802	50.1	1.038 ± 0.042	7.276
3X151	sanidine	6.89 6.95	7.498 8.788	68.6 71.4	0.9662 ± 0.0676 1.126 ± 0.045	5.986
3X170	sanidine	7.22 7.25	9.926 7.741	75.1 73.4	1.117 ± 0.034 1.146 ± 0.046	6.194
4D014	sanidine	7.25 7.30	10.464 7.902	78.7 60.9	1.117 ± 0.034 1.236 ± 0.043	6.403
3X177	sanidine	7.12 7.14	8.513	78.8	1.561 ± 0.094	8.665
3X146	sanidine	6.64 6.82	7.890 7.909	64.0 76.7	1.372 ± 0.055 1.490 ± 0.060	8.419
4D015	sanidine	7.00 7.00	9.188 7.237	84.2 64.3	1.386 ± 0.042 1.440 ± 0.072	7.990

*The ± figure is an estimate of the standard deviation of precision. See text for detailed discussion.

†Samples from the altered zone of the Valles Caldera. See text for details.

found by Dalrymple and Hirooka (1965) for a 3.4 million-year-old basalt. This difference is not surprising, however, because analyses on younger material usually involve smaller total amounts of radiogenic argon per analysis and have higher atmospheric corrections, so that precision is expected to be less than on old rocks.

To obtain the estimated standard deviation of precision for the calculated ages (Table 3), the values of s for the argon analyses were combined, by appropriate statistical methods, with the average precision figure of the K_2O analyses, which is 0.5 percent.

A completely independent estimate of precision can be obtained from a statistical analysis of the replicate measurement using methods described by Crow and others (1960) and McIntyre (1963). This gives a pooled estimate of the standard deviation of precision, s_o, expressed in percent. In order to

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TABLE 3. POTASSIUM-ARGON AGES OF ROCKS FROM THE VALLES CALDERA, NEW MEXICO
UNITS ARE LISTED IN STRATIGRAPHIC ORDER WHERE KNOWN

	Name and Unit No.		Sample No.	Material	Calculated Age* (10^6 years)
Late rhyolite — Group C	San Antonio II	S45	4D069	sanidine	0.434 ± 0.015
	South Mountain	S44	3X074	sanidine	0.494 ± 0.015
	La Jara	S43	3X203	sanidine	0.502 ± 0.015
	San Antonio I	S42	3X062	sanidine	0.535 ± 0.015
Late rhyolite — Group B	San Luis	S41	3X122	sanidine	0.823 ± 0.074
			4D214	sanidine	0.692 ± 0.015
	Santa Rosa II	S40	4D003	sanidine	0.707 ± 0.019
	Seco	S39	4D001	sanidine	0.726 ± 0.015
	Santa Rosa I	S38	4D002	sanidine	0.884 ± 0.028
	Abrigo III	S37	3X189	sanidine	0.886 ± 0.019
	Warm Springs	S36	3X117†	sanidine	1.22 ± 0.04
Late rhyolite — Group A	Del Medio I	S35	4D005	obsidian	1.04 ± 0.05
	Del Medio II	S34	4D004	obsidian	1.15 ± 0.03
Middle rhyolite	Middle rhyolite	S33	4D207†	biotite	1.71 ± 0.07
			4D013†	sanidine	1.28 ± 0.06
				biotite	1.60 ± 0.11
			3X114†	plagioclase	3.25 ± 0.32
Early rhyolite	Early rhyolite	S32	3X110†	sanidine	1.25 ± 0.11
Bandelier Tuff — Upper member	Anorthoclase unit		3X241	anorthoclase	1.19 ± 0.04
			5A057†	anorthoclase	1.24 ± 0.05
	Site 5	S31	3X151	sanidine	1.02 ± 0.04
	Site 2		3X170	sanidine	1.06 ± 0.03
	Basal pumice fall		4D014	sanidine	1.09 ± 0.03
Bandelier Tuff — Lower member	Site 6		3X177	sanidine	1.48 ± 0.09
	Site 1	S30	3X146	sanidine	1.44 ± 0.04
	Basal pumice fall		4D015	sanidine	1.37 ± 0.04

*The ± figure is an estimate of the standard deviation of precision. See text for detailed discussion. $\lambda_\epsilon = 0.585 \times 10^{-10}$ yr^{-1}, $\lambda_\beta = 4.72 \times 10^{-10}$ yr^{-1}, $K^{40}/K_{total} = 1.19 \times 10^{-4}$ mole/mole.

†Samples from the altered zone of the Valles Caldera. See text for details.

minimize the possible effect on the statistics of errors introduced by geological factors, such as alteration and contamination, data on units from the altered zone or from the welded or crystallized portions of the Bandelier Tuff have not been used. However, had all data been included, the statistics would not be changed substantially.

For the results in Table 2, s_o is 0.51 percent for the potassium analyses, 3.95 percent for Ar^{40}_{rad}/gm^{-1}, and 3.98 percent for Ar^{40}_{rad}/K^{40}. From these data, it can be inferred that the standard deviation of precision, σ, for the popula-

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tion of all calculated ages in the range of from 0.4 to 1.4 m.y. and of similar experimental quality, is 3.1 to 5.7 percent with 90 percent confidence (Crow and others, 1960, Table 8). This pooled estimate of the precision is in good agreement with the average value of 4.5 percent estimated previously, and suggests that the precision values given in Tables 2 and 3 are probably reasonable estimates.

For the present study it is useful to know the maximum confidence, P, with which a real difference between any two apparent ages can be claimed. Values of P for 13 of the rock units studied have been calculated (Fig. 11) using the probability density function for a normal curve and the precision estimates in Table 3. Large values of P indicate that the analyses are precise enough to resolve the indicated difference in calculated age; small values of P indicate that they are not. However, it should be emphasized that the latter does not establish that the ages are the same, only that the data do not allow us to claim that they are different. To illustrate the use of this table, P is 97 percent for the difference between the calculated ages of South Mountain (0.49 m.y.) and San Antonio Mountain I (0.54 m.y.) and 46 percent for the difference in the calculated ages of Cerro San Luis (0.69 m.y.) and Cerro Santa Rosa II (0.71 m.y.). At the usual 95 percent confidence level, the difference in calculated ages for the first pair is significant, and the difference in calculated ages for the second pair is not.

Discussion of the Age Determinations

Figure 11 shows the most probable potassium-argon ages of the units studied. The units from the altered zone have been omitted for reasons discussed below.

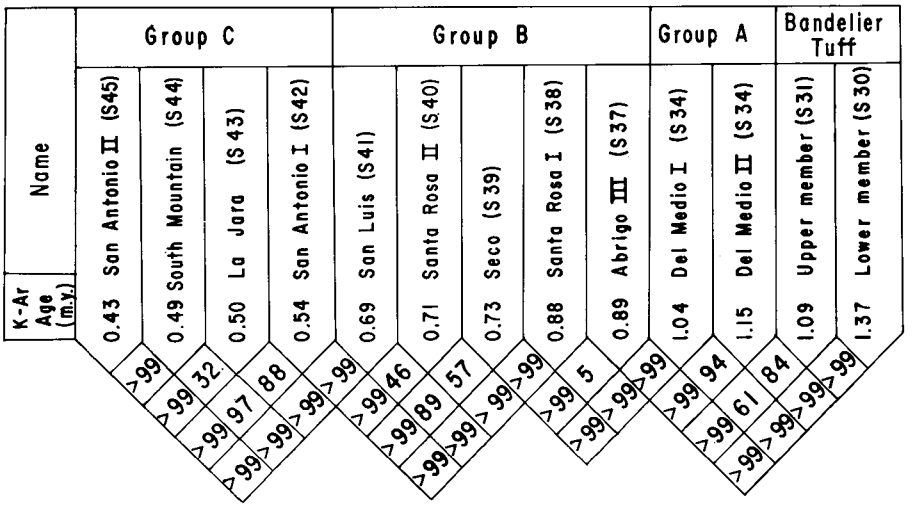


Figure 11. Maximum confidence, P in percent, that the calculated ages of any two units are truly different and not just due to random errors in the analytical procedure.

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For the age of Cerro San Luis (S41), the age of sample 4D214 has been used rather than that of 3X122 because only one argon determination was made on 3X122, whereas the age of 4D214 is the result of four argon measurements. The age of 0.72 given for this unit in a previous publication (Doell and Dalrymple, 1966) was the mean of all five measurements.

For the Bandelier Tuff, the calculated ages of the pumice fragments from the basal ash falls are considered to be the more reliable indicators of the ages of the lower and upper members, and these values have been used in Figure 11. However, the dating of ash-flow tuffs is complicated by several factors, which deserve discussion.

It is well recognized that ash-flow tuffs normally contain foreign rock fragments. Ross and Smith (1961, p. 35), in studying hundreds of thin sections of welded tuffs of many ages and from many parts of the world, found that inclusions are rarely, if ever, absent. Much of the included material is fine grained and is impossible to completely separate from whole-rock samples. Dalrymple and others (1965), in their study of the Bishop Tuff, showed that this contamination by inherited argon can cause potassium-argon ages to be too old.

The Bandelier Tuff, no exception to the problem of contamination, contains inclusions of most of the older rocks in the area. The distribution of these inclusions indicates that most were picked up during eruption and emplacement of the deposits and not incorporated in the magma prior to eruption; hence, pumice blocks offer the cleanest samples for age studies, but they are difficult to isolate in any but the nonwelded, vitric zones of the Bandelier Tuff.

A second factor of possible importance to the accuracy of the K-Ar age determinations on the Bandelier Tuff is the unmixing of sanidine, which occurs in varying degree throughout most of the unit. Although quantitative measurement of the degree of unmixing has not yet been possible, many X-ray measurements of sanidine phenocrysts show a progressive change from a homogeneous state, which is found only in the quickly "quenched" air falls and chilled boundaries of the ash flows, to an unmixed state, which clearly shows two phases, in the interior of the cooling units. All unmixed feldspars so far examined in the Bandelier are cryptoperthites; no microperthites have been observed.

Sanidine phenocrysts in some portions of vapor-phase zones show overgrowths of sanidine. This is another factor of unassessed importance.

Considering both the problem of contamination and the variables related to the crystallization histories of the cooling units, it would seem that the feldspars from the quenched air-fall pumice would be the most reliable for accurate age determination. However, a third factor of unknown significance concerns the crystallization site of the feldspar in the magma chamber.

Chemical studies of the upper member (Smith and Bailey, 1966) indicate that the first erupted pumices, which are inferred to have come from the top of the chamber, contain a measurably higher concentration of volatility than

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later pumice. The feldspars of the air-fall pumice contain many fluid and glass inclusion sites, and, therefore, argon also may have been concentrated in the upper part of the chamber. It seems still a question as to what sample is most significant for the most accurate age determination in deposits like the Bandelier Tuff. However, because of the proven effects of contamination by older material in ash-flow tuffs, the air-fall ages are considered to be the more reliable and these are used in Figure 11.

Sample sites of the Bandelier Tuff for the present study were selected to span the total stratigraphic assemblages of each member (Pl. 1; Fig. 9). They were also taken from a variety of lithologies that are representative of the different regimes of temperature, pressure, and cooling rates that govern the final welded, crystallized, or quenched state of a given body of tuff.

Within the lower member of the Bandelier Tuff, a difference in any of the three calculated ages cannot be claimed with 90 percent confidence. This is also true for the upper member if the indicated age of the anorthoclase unit is neglected. The probability of a real difference in calculated age between the anorthoclase unit and any of the other three dated units within the upper member is greater than 94 percent. The cause of this anomalous age is unknown, but it may be due to one of the factors discussed above.

The ages obtained for units in the altered zone of the caldera are puzzling. They seem to be generally older than is permitted by the known stratigraphy and the ages of underlying units. The indicated age of Warm Springs dome is greater at the 90 percent confidence level than that of Del Medio I and is considerably older than the other domes in group B. The ages obtained on samples from the middle rhyolite are not only significantly older than the ages of the Bandelier Tuff, but are internally discordant.

Only one unit sampled, the anorthoclase unit, occurs both within and without the altered zone. Samples of this rock from both places were analyzed to see whether or not the tendency for older ages from the altered zone could be directly related to the alteration. However, because the apparent ages of both samples are not different at even the 60 percent confidence level, the test can be interpreted only as inconclusive.

Only four mechanisms that could cause the anomalously old ages from the altered zone seem possible: (1) potassium loss by base exchange during alteration, (2) argon gain during alteration, (3) introduction of excess argon into the minerals before or during emplacement, and (4) inherited argon (contamination) caused by the incorporation of older material during emplacement. Mechanism (1) seems unlikely without an accompanying loss of argon. Kulp and Engels (1963) found in laboratory experiments that as much as 50 percent of the K in a metamorphic biotite could be removed by base exchange without any significant effect on the potassium-argon age. Further removal of potassium resulted in argon loss and lower ages due to extreme lattice disordering. However, it may be inappropriate to extrapolate these biotite results to feldspars. Evidence concerning (2) has never been reported, and it is difficult to see how it might operate at the low pressures under which

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alteration occurred. It is difficult to see why either excess argon (3) or contamination (4) would be seemingly prevalent in the altered zone. In addition, xenoliths or xenocrysts have not been observed in the Warm Springs dome or in the middle or early rhyolites. From the present data, there is no reason to either preclude or favor any of the four possibilities, and the ages obtained on samples from the altered zone are not considered reliable indicators of the age of these units, nor, as stated above, do samples from these same units yield meaningful paleomagnetic results. Thus, it does appear that the anomalous ages and ambiguous magnetic properties are connected in some way with the alteration.

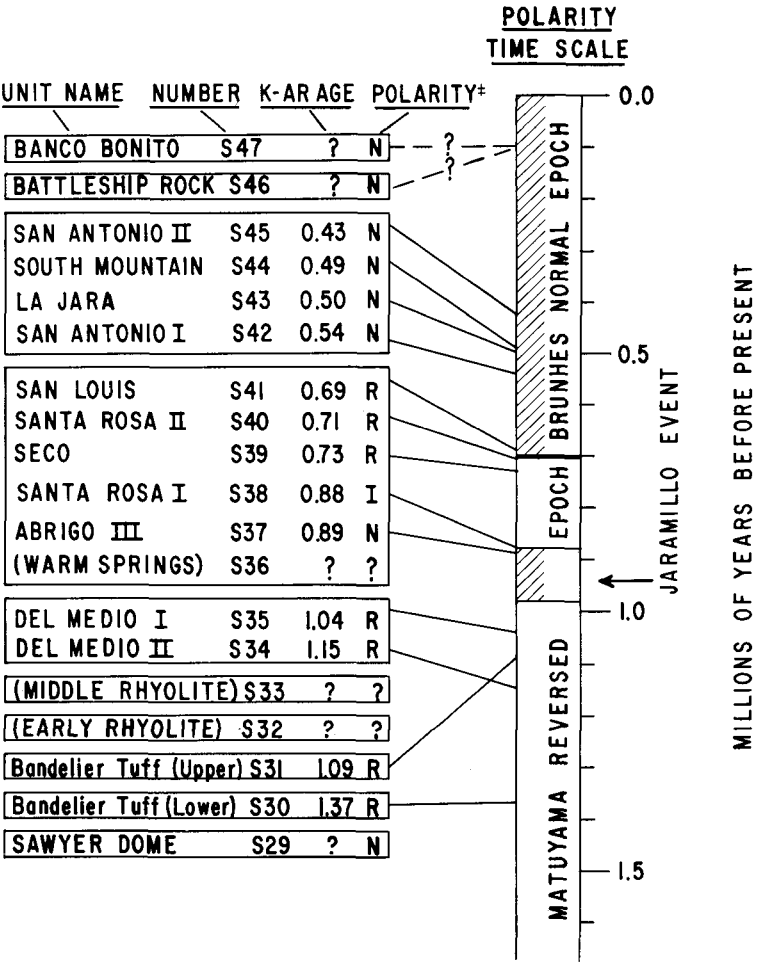
All of the ages given in Figure 11 agree, within the estimated experimental limits of the method at the 90 percent confidence level, with the known stratigraphic relationships. In addition, the values of P for these units suggest that it is possible to distinguish between most of the domes within a given group.

CONCLUSIONS

The stratigraphy of the Valles caldera igneous rocks, their magnetic polarity, the best estimate of their potassium-argon ages, and the relationship of these data to the previously determined geomagnetic polarity time scale for the past 1.5 m.y. (Cox and others, 1964; Dalrymple and others, 1965; Doell and others, 1966; Doell and Dalrymple, 1966; McDougall and Chamalaun, 1966) are summarized in Figure 12. Except for the one discrepancy between the stratigraphy and potassium-argon ages of Cerro del Medio II and the Bandelier's upper member (significant only at the 84 percent level) and the data from the altered zone that are excluded for the reasons stated above, these data are in excellent agreement, and there is little doubt that the summary represents the history of emplacement of these units. From the time of extrusion of the lower member of the Bandelier Tuff, there has been more or less continuous volcanic activity in this region, often of an explosive nature, and there is certainly no reason to expect that this igneous activity is, geologically speaking, less intense at present than it was 1.5 m.y. ago. Even though potassium-argon ages were not obtained for the two most recent events (Battleship Rock and Banco Bonito flows), they clearly took place as recently as about a hundred thousand years ago (Smith and Bailey, 1968), and Figure 12 suggests a "periodicity" for volcanism in this region which does not preclude future eruptions.

As mentioned earlier, stratigraphic relations indicate that Sawyer dome is older than the Bandelier's upper member, but its relationship to the lower member was not known. Although no radiometric age was obtained, the paleomagnetic results that yield a normal polarity indicate that it is also older than the lower member, unless there is another normal polarity event between 1.0 and 1.5 m.y. ago. This is rather unlikely from previous geomagnetic polarity studies, and the next older known period of normal polarity is the

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‡PLACED IN STRATIGRAPHIC SUCCESSION (STRATIGRAPHIC RELATIONS BETWEEN UNITS IN A GIVEN BOX IS UNKNOWN)

Figure 12. Relationships of stratigraphy, K-Ar ages, and magnetic directions of the rhyolites and associated rocks from the Valles Caldera, New Mexico. Polarity: N-normal, R-reversed, I-intermediate.

Olduvai normal event near 1.9 m.y. ago, or possibly the proposed Gilsá event near 1.6 m.y. ago (McDougall and Wensink, 1966).

Doell and Dalrymple (1966), in a preliminary report, used the age-polarity data for the rhyolites of Cerro Santa Rosa II, Cerro San Luis, Cerro Seco, Cerro Santa Rosa I, Cerro Abrigo III, and Cerro del Medio I for defining the Jaramillo event (named after Jaramillo Creek — which flows east and south of these domes). Since then, new data that confirm this event have been reported by McDougall and Chamalaun (1966) from Victoria, Australia,

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and Réunion Island, Indian Ocean, and by Hoare and others (1968) from Nunivak Island, Alaska. The event recently has also been noted in magnetic studies of deep-sea cores (Opdyke and others, 1966; Ninkovich and others, 1966) and in midoceanic-ridge magnetic-anomaly patterns that are interpreted in terms of the "ocean-floor spreading" hypothesis (Vine, 1966). Thus, there is now little doubt concerning this event. Moreover, it appears that the intermediate direction that is observed in Cerro Santa Rosa I marks the termination of the Jaramillo event rather than the beginning (McDougall and Chamalaun, 1966; Ninkovich and others, 1966) and that the event was of rather short duration — about 50,000 years (Ninkovich and others, 1966).

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REFERENCES CITED

Cox, Allan, 1961, Anomalous remanent magnetization of basalt: U.S. Geol. Survey Bull. 1083–E, p. 131–160.

Cox, Allan, and Doell, R. R., 1960, Review of paleomagnetism: Geol. Soc. America Bull., v. 71, p. 645–768.

—— 1964, Long period variations of the geomagnetic field: Seismol. Soc. America Bull., v. 54, (no. 6, pt. B, Byerly Volume) p. 2243–2270.

—— 1967, Measurement of high coercivity magnetic anisotropy, *in* Collinson, D. W., Creer, K. W., and Runcorn, S. K., *Editors,* Developments in solid earth geophysics 3, Methods in paleomagnetism: Amsterdam, Elsevier Pub. Co., p. 477–482.

Cox, Allan, and Dalrymple, G. B., 1967, Statistical analysis of geomagnetic reversal data and the precision of potassium-argon dating: Jour. Geophys. Research, v. 72, p. 2603–2614.

Cox, Allan, Doell, R. R., and Dalrymple, G. B., 1963, Geomagnetic polarity epochs — Sierra Nevada II: Science, v. 142, p. 382–385.

—— 1964, Reversals of the earth's magnetic field: Science, v. 144, p. 1537–1543.

Crow, E. L., Davis, F. A., and Maxfield, M. W., 1960, Statistics manual: New York, Dover Publications, Inc., 288 p.

Dalrymple, G. B., Cox, Allan, and Doell, R. R., 1965, Potassium-argon age and paleomagnetism of the Bishop Tuff, California: Geol. Soc. America Bull., v. 76, p. 665–673.

Dalrymple, G. B., and Hirooka, Kimio, 1965, Variation of potassium, argon, and calculated age in a late Cenozoic basalt: Jour. Geophys. Research, v. 70, p. 5291–5296.

Downloaded from memoirs.gsapubs.org on January 18, 2014

Doell, R. R., and Cox, Allan, 1963, The accuracy of the paleomagnetic method as evaluated from historic Hawaiian lava flows: Jour. Geophys. Research, v. 68, p. 1997–2009.

—— 1967a, Analysis of alternating-field demagnetization equipment, *in* Collinson, D. W., Creer, K. M., and Runcorn, S. K., *Editors,* Developments in solid earth geophysics 3, Methods in paleomagnetism: Amsterdam, Elsevier Pub. Co., p. 241–253.

—— 1967b, A spinner magnetometer for igneous rocks, *in* Collinson, D. W., Creer, K. M., and Runcorn, S. K., *Editors,* Developments in solid earth geophysics 3, Methods in paleomagnetism: Amsterdam, Elsevier Pub. Co., p. 136–141.

—— 1967c, Measurement of natural remanent magnetization at the outcrop, *in* Collinson, D. W., Creer, K. M., and Runcorn, S. K., *Editors,* Developments in solid earth geophysics 3, Methods in paleomagnetism: Amsterdam, Elsevier Pub. Co., p. 159–162.

—— 1967d, Paleomagnetic sampling with a portable coring drill, *in* Collinson, D. W., Creer, K. M., and Runcorn, S. K., *Editors,* Developments in solid earth geophysics 3, Methods in paleomagnetism: Amsterdam, Elsevier Pub. Co., p. 21–25.

—— 1967e, Recording magnetic balance, *in* Collinson, D. W., Creer, K. M., and Runcorn, S. K., *Editors,* Developments in solid earth geophysics 3, Methods in paleomagnetism: Amsterdam, Elsevier Pub. Co., p. 440–444.

Doell, R. R. and Dalrymple, G. B., 1966, Geomagnetic polarity epochs — A new polarity event and the age of the Brunhes-Matuyama boundary: Science, v. 152, p. 1060–1061.

Doell, R. R., Dalrymple, G. B., and Cox, Allan, 1966, Geomagnetic polarity epochs — Sierra Nevada data, 3: Jour. Geophys. Research, v. 71, p. 531–541.

Evernden, J. F., and Curtis, G. H., 1965, The potassium-argon dating of late Cenozoic rocks in East Africa and Italy: Current Anthropology, v. 6, p. 343-385.

Fisher, R. A., 1953, Dispersion on a sphere: Royal Soc. [London] Proc., ser. A, v. 217, p. 295–305.

Graham, K.W.T., 1961, The re-magnetization of a surface outcrop by lightning currents: Royal Astron. Soc. Geophys. Jour., v. 6, p. 85–102.

Griggs, R. L., 1964, Geology and ground water resources of the Los Alamos area, New Mexico: U.S. Geol. Survey Water-Supply Paper 1753, 107 p.

Hoare, J. M., Condon, W. H., Cox, Allan, and Dalrymple, G. B., 1968, Geology, paleomagnetism, and potassium-argon ages of volcanic rocks from Nunivak Island, Alaska: Geol. Soc. America Mem. 116, p. 377–413.

Irving, E., Stott, P. M., and Ward, M. A., 1961, Demagnetization of igneous rocks by alternating magnetic fields: Philos. Mag., v. 6, p. 225–241.

Kulp, J. L., and Engels, Joan, 1963, Discordances in K-Ar and Rb-Sr isotopic ages, *in* Proc. Internat. Atomic Energy Symposium on Radioactive Dating, Athens, 19–23 Nov., 1962, p. 219–238.

Lanphere, M. A., and Dalrymple, G. B., 1965, P–207 — An interlaboratory standard muscovite for argon and potassium analyses: Jour. Geophys. Research, v. 70, p. 3497–3503.

—— 1966, Simplified bulb-tracer system for argon analyses: Nature, v. 209, p. 902–903.

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McDougall, Ian, and Chamalaun, F. H., 1966, Geomagnetic polarity scale of time: Nature, v. 212, p. 1415–1418.

McDougall, Ian, and Wensink, H., 1966, Paleomagnetism and geochronology of the Pliocene-Pleistocene lavas in Iceland: Earth and Planetary Sci. Letters, v. 1, p. 232–236.

McIntyre, D. B., 1963, Precision and resolution in geochronometry, *in* Albritton, Claude C., *Editor,* The Fabric of Geology: Reading, Mass., Addison-Wesley Pub. Co., Inc., p. 112–139.

Ninkovich, Dragoslav, Opdyke, Neil, Heezen, B. C., and Foster, J. H., 1966, Paleomagnetic stratigraphy, rates of deposition and tephrachronology in north Pacific deep-sea sediments: Earth and Planetary Sci. Letters, 1, p. 476–492.

Opdyke, N. D., Glass, B., Hays, J. D., and Foster, J. H., 1966, Paleomagnetic study of Antarctic deep-sea cores: Science, v. 154, p. 349–357.

Rimbert, F., 1956, Sur l'action de champs alternatifs sur des roches portant une aimantation remanente isotherme de viscosite: Acad. Sci. [Paris] Comptes Rendus, v. 242, p. 2536–2538.

Ross, C. S., and Smith, R. L., 1961, Ash flow tuffs, their origin, geologic relations, and identification: U.S. Geol. Survey Prof. Paper 366, 81 p.

Ross, C. S., Smith, R. L., and Bailey, R. A., 1963, Outline of the geology of the Jemez Mountains, New Mexico: Guidebook New Mexico Geol. Soc. Twelfth Field Conf., Albuquerque, N. M., 1961, p. 138–145.

Shapiro, Leonard, and Brannock, W. W., 1962, Rapid analysis of silicate, carbonate, and phosphate rocks: U.S. Geol. Survey Bull. 1144–A, p. A1–A56.

Smith, R. L., 1960, Zones and zonal variations in welded ash flows: U.S. Geol. Survey Prof. Paper 345–F, p. 149–159.

Smith, R. L., Bailey, R. D., and Ross, C. S., in press, Geologic map of the Jemez Mountains, New Mexico: U.S. Geol. Survey Misc. Geol. Inv. Map I-571.

Smith, R. L., and Bailey, R. A., 1966, The Bandelier Tuff — A study of ash-flow eruption cycles from zoned magma chambers: Bull. Volcanol. ser. 2, v. 29, p. 83–104.

——— 1968, Resurgent cauldrons: Geol. Soc. America Mem. 116, p. 613–662.

Smith, R. L., Bailey, R. A., and Ross, C. S., 1961, Structural evolution of the Valles Caldera, New Mexico, and its bearing on the emplacement of ring dikes: U.S. Geol. Survey Prof. Paper 424–D, p. 145–149.

Uyeda, Seiya, 1958, Thermo-remanent magnetism as a medium of paleomagnetism, with special reference to reverse thermo-remanent magnetism: Japanese Jour. Geophysics, v. 2, p. 1–123.

Uyeda, Seiya, Fuller, M. D., Belshe, J. C., and Girdler, R. W., 1963, Anisotropy of magnetic susceptibility of rocks and minerals: Jour. Geophys. Research, v. 68, p. 279–291.

Verhoogen, John, 1962, Oxidation of iron-titanium oxides in igneous rocks: Jour. Geology, v. 70, p. 168–181.

Vine, F. J., 1966, Spreading of the ocean floor — New evidence: Science, v. 154, p. 1405–1415.

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APPENDIX

Sample Locations and Descriptions

Description of sample sites (unless otherwise noted, all potassium-argon samples were collected within the span of paleomagnetic samples at the same exposures). Map source: U.S. Geol. Survey topographic quadrangle maps: Jemez Springs quadrangle, New Mexico, 1952, scale 1:62,500. Frijoles quadrangle, New Mexico, 1953, scale 1:62,500. White Rock quadrangle, New Mexico, 1952, scale 1:24,000.

Banco Bonito (S47): Northeast of Jemez River Road, 2 miles north of Battleship Rock. Four cores 1500 feet S. 50° E., BM 7634; four cores 400 feet N. 80° E., BM 7565. Center Jemez Springs quadrangle.

Battleship Rock (S46): Six cores from northwest and southeast sides of Battleship Rock, 900 feet S. 75° E., BM 6755; three cores near La Cueva, 500 feet N. 40° W., BM 7634, 7700-foot elevation. Center Jemez Springs quadrangle.

San Antonio II (S45): East side San Antonio Mountain, 3400 feet N. 29° W., BM 8653, 9250-foot elevation. North-center Jemez Springs quadrangle.

South Mountain (S44): North side La Cueva-Otowi Road, three cores at BM 7937, five cores at BM 8407, and three cores 2900 feet S. 83° W., elevation point 8761, 8600-foot elevation. SE¼ Jemez Springs quadrangle.

La Jara (S43): South side Cerro La Jara, 500 meters east of west quadrangle boundary, 8560-foot elevation, 1950 m S. 20° W., BM 8529. West center Frijoles quadrangle.

San Antonio I dome (S42): Three cores from east side Water Canyon, 4800 feet S., 67 W., BM 8062, 8350-foot elevation, two cores 2100 feet N. 52° E. at elevation point 8830 at 8725-foot elevation, and three cores from north side Ice Cave Canyon, 2100 feet S. 82° E. of junction of Ice Cave Canyon with San Antonio Creek, 8500-foot elevation. Center Jemez Springs quadrangle.

San Luis dome (S41): Four cores southwest side hill 9504, 2200 feet N. 52° W., BM 8702, 8750-foot elevation; eight cores east-southeast side, 2000 feet N. 7° E., BM 8727, 8900-foot elevation. Northeast corner Jemez Springs quadrangle.

Santa Rosa II dome (S40): 870 meters S. 48° W., BM 8555, south side Cerro Santa Rosa, 8960-foot elevation. Northwest corner Frijoles quadrangle.

Seco dome (S39): East side hill 9951, 2900 feet N. 2° E., BM 8787, 9000-foot elevation. Northeast corner Jemez Springs quadrangle.

Santa Rosa I dome (S38): 400 meters east of west quadrangle boundary, 2000 meters S. 10° W. from BM 8555, 9100-foot elevation. Northwest corner Frijoles quadrangle.

Abrigo III dome (S37): Northwest side of Cerro del Abrigo, 2600 feet S. 31° E. BM 8555 at 8700-foot elevation. Northwest corner Frijoles quadrangle.

Warm Springs dome (S36): 1700 feet S. 40° W. of elevation point 8496 in Valle San Antonio at 8500-foot elevation. Northeast corner of Jemez Springs quadrangle.

Del Medio I dome (S35): West side of Cerro del Medio, 800 meters N. 81° E., BM 8701, 8750-foot elevation. NW¼ Frijoles quadrangle.

Del Medio II dome (S34): Near Puerto de Abrigo, northwest side of Cerro del Medio, 2980 meters, N. 79° E., BM 8739, 8940-foot elevation. NW¼ Frijoles quadrangle.

Middle rhyolite (S33): 17 cores northwest of San Antonio Mountain, 3200 feet S. 51° W. forest boundary marker 8488 feet at 8750-foot elevation, eight cores east of Water Canyon, 1700 feet N. 55° W., BM 7874, at 8250-foot elevation; 11 cores northeast source of Redondo Creek, 3700 feet S. 81° E., hill 9914, at 9500-foot elevation. North half of Jemez Springs quadrangle.

Early rhyolite (S32): South of Deer Canyon, 7800 feet N. 88° E., BM 7734, at 9000-foot elevation. Center Jemez Springs quadrangle.

Upper member of the Bandelier Tuff (S31): (1) Anorthoclase unit west portion Ramon Vigil Grant, 5000 feet S. 88° E., BM 7628, Frijoles quadrangle; dated sample from altered zone (sample 5A057) from south end of Redondo Border approximately 1400 feet northeast of elevation 8618, Jemez Springs quadrangle. (2) 4 levels (sites 2-5) west side of Virgin Canyon 3200 feet east and 5200 feet north of the southwest corner of the Jemez Springs quadrangle. (3) Site 1 west of the Ramon Vigil Grant, 1800 feet S. 78° W. from BM 7628, Frijoles quadrangle. (4) Basal pumice fall on south side of Highway 4, 200 meters due east of northwest corner sec. 20, T. 19 N., R. 7 E., at 6630-foot elevation, White Rock quadrangle.

Lower member of the Bandelier Tuff (S30): Four levels (sites 3-6) SW¼ sec. 26, T. 20 N., R. 2 E., 2200 feet S. 78° W. Seven Springs Ranger Station, northwest of Jemez Springs quadrangle. (2) Two levels (sites 1 and 2) northwest of Battleship Rock, 3500 feet N. 73° W. BM 6943, Jemez Springs quadrangle. (3) Basal pumice fall on north side of Highway 4, 230 meters S. 35° W. from NE cor. sec. 21, T. 19 N., R. 7 E., at 6240-foot elevation, White Rock quadrangle.

Sawyer dome (S29): South of Valle Grande, 4700 feet N. 66° W., BM 8949, 9300-foot elevation. West-center Frijoles quadrangle.